

Credit Card Customer NPV Estimation Component: A Full Evidence-Integrated Proposal for a Replacement Valuation Module

Chak Wong

July 1, 2026

Abstract

This report proposes a replacement net present value (NPV) estimation component for credit-card acquisition, cross-sell, line, offer, and account-management decisions at a large U.S. retail bank. The component is not a campaign platform, underwriting engine, finance ledger, stress-testing system, or reporting platform. It is the governed valuation module that those existing systems call when they need an estimate of incremental risk-adjusted customer value.

The proposal is intentionally full-depth. It integrates the existing literature survey into the component design rather than replacing the survey with a short memo. The literature body is preserved because it supplies the evidence for the component's behavioral and financial mechanisms: household-finance and credit-card studies show that employment, liquidity, credit limits, interest rates, transfers, product terms, and macro shocks affect spending, borrowing, payments, and losses; payment-choice and rewards studies show that card adoption and card use are distinct outcomes; customer lifetime value and duration models support lifetime activity, dormancy, and attrition modules; and credit-risk, accounting, and supervisory sources support decomposing value into balances, PPNR, expected loss, funding, capital, costs, and scenarios. The proposal then translates that evidence into a practical component specification: decision-context semantics, valuation-policy identifiers, request and response contracts, status handling, operating modes, identification discipline, external and internal data strategy, validation, migration, governance, and source-support limits. Public literature supports mechanisms and model structure; public and licensed data support scenarios, controls, benchmarks, and priors; issuer-specific activation, cannibalization, spend response, loss elasticity, and NPV parameters must be estimated from the bank's own data, experiments, quasi-experiments, and reconciliations.

1 Executive Summary and Component Charter

The bank's immediate task is not to build a new decisioning platform. Marketing systems will still choose and execute campaigns; underwriting systems will still apply credit policy; account-management systems will still execute line, retention, conversion, and closure treatments; finance systems will still own ledgers, planning, and official reporting; and stress-testing systems will still own supervisory stress processes. The proposed work is narrower and more precise: replace the current credit-card customer value component with a governed NPV estimator that can be called by those existing systems.

The component charter is:

The NPV component owns valuation. Caller systems own action selection, policy thresholds, campaign orchestration, underwriting rules, line-management workflow, finance reporting, and customer execution.

The component should answer three questions for every call:

1. **What decision context is being valued?** Targeting, existing client cross-sell, underwriting, line assignment, offer choice, product conversion, retention, and portfolio analysis have different populations, data availability, action sets, and counterfactuals.
2. **What valuation semantics and assumption bundle are being used?** A number called “NPV” is not useful unless horizon, discounting, funding, capital, tax, rewards, costs, terminal value, relationship value, cannibalization, macro scenario, and downstream policy assumptions are explicit and versioned.
3. **What can the caller safely do with the output?** The component may return values, decompositions, warnings, statuses, and non-binding option rankings. It should not itself mail a customer, approve an application, choose a line, close an account, or book a finance result.

The central valuation object is incremental value:

$$\Delta\text{NPV}_i(a; d, s, \pi) = \mathbb{E}[\text{NPV}_i(a; d, s, \pi) - \text{NPV}_i(a_0; d, s, \pi) \mid \mathcal{I}_{id}], \quad (1)$$

where i is the customer, prospect, account, household, application, or account-action pair being scored; a is a candidate action; a_0 is the decision-specific baseline; d is the decision context; s is the scenario and valuation-assumption bundle; π is the assumed downstream policy; and $\mathcal{I}_{id}$ is the information legally and operationally available at that decision point.

This definition prevents several common errors. A high response score is not necessarily high value. A high first-spend account may cannibalize an old bank card. A profitable-looking high-line decision may create exposure, capital, and losses that offset additional interchange or finance charges. A retention or product-conversion action may preserve value that would otherwise attrite, but it may also give away rewards or pricing concessions to customers who would have stayed anyway. For this reason, the component should return a tagged and decomposed value object, not a bare scalar score.

The component must also treat card wallet as a partially observed state, not as a directly observed label. The bank observes its own new-card transactions, old-card transactions, deposits, debit activity, balances, payments, fees, and losses. It usually observes competitor-card spend, competitor revolving balances, card-on-file position, and outside payment routing only through partial signals such as bureau attributes, balance-transfer records, payoff records, merchant or network aggregates, account-aggregation samples, or campaign experiments. Therefore the core modeling question is not just “how much will the customer spend on the new card?” It is how the action changes the allocation of total spend, revolving balances, payment behavior, losses, rewards cost, and relationship activity across the bank’s old card, the bank’s new card, debit or deposit rails, and competitors. This distinction separates outside wallet size, movable wallet, causal response, and NPV.

The recommended first production slice is intentionally narrow: one controlled marketing or prescreen acquisition consumer, batch scoring, one offer family, and one incremental pre-tax valuation semantics bundle. This permits frozen input snapshots, campaign holdouts, shadow scoring against the current component, portfolio-level reconciliation, and no real-time underwriting dependency. Expansion to underwriting, line assignment, existing-account treatments, finance planning, and account-management use cases should require separate use-case approval.

The rest of the proposal should be read through five hard facts about card NPV. First, NPV is latent and assumption-sensitive: it depends on long-horizon spend, revolve, attrition, loss, funding, capital, cost, and terminal-value assumptions that are only partially observed. Second, card value

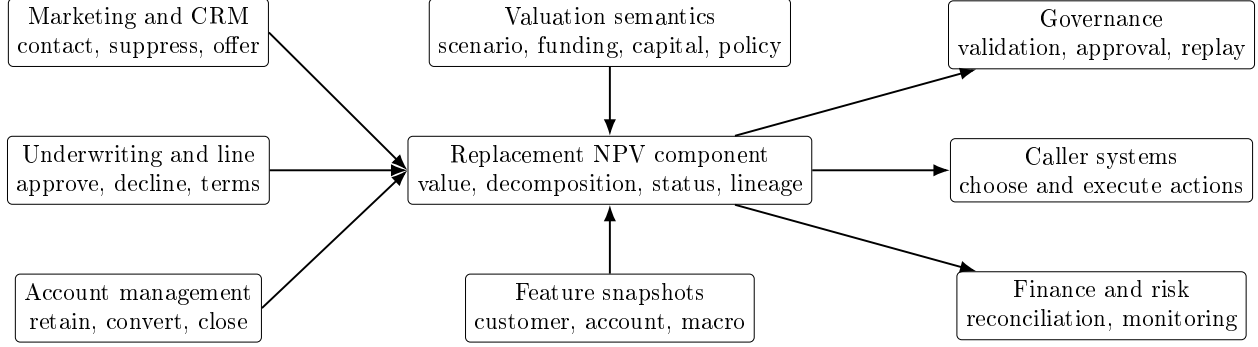


Figure 1: System context: the proposal replaces the valuation component, not the caller systems.

is behaviorally heterogeneous: spend, revolve, utilization, payment behavior, rewards response, attrition, losses, funding, and capital interact rather than collapsing to one response or risk score. Third, the scored population changes through the funnel from prospect to applicant to approved account to booked and active account, so the available data and valid estimand change. Fourth, the objective changes by decision: contact-versus-suppress, approve-versus-decline, line-option choice, product conversion, and retention are different counterfactuals. Fifth, decisions are sequentially interdependent: upstream acquisition and approval choices change downstream account populations, while future line, pricing, retention, collections, and conversion policies change the economics assumed upstream. These facts are the reason the component needs decision-context declarations, valuation-semantics identifiers, decomposed outputs, statuses, lineage, and use-case-specific validation.

2 Replacement Boundary and How the Evidence Is Used

The current component should be inventoried before replacement. The bank should identify all current consumers, input fields, output fields, thresholds, fallbacks, latency classes, owners, model versions, valuation assumptions, manual overrides, and downstream reports that depend on the current value measure. This proposal does not assume those facts are known. It treats them as discovery items that must be resolved before cutover.

Typical weaknesses to test in the current component are:

- response, approval, activation, or sales volume used as value proxies;
- unclear counterfactuals for contact, approval, line, retention, or product-conversion decisions;
- weak cannibalization treatment for existing-client new-card offers;
- one value definition reused across different funnel stages;
- missing decomposition into spend, balances, payments, PPNR, losses, funding, capital, rewards, fraud, servicing, and terminal value;
- stale discount, funding, capital, tax, cost, or reward assumptions;
- output fields that are difficult to reconcile to finance, risk, or model-risk views.

This report uses four evidence classes. First, **literature evidence** supports mechanisms and modeling structures: income shocks affect spend, payment instruments have adoption and use stages, rewards can change usage, CLV models represent lifetime activity, and causal methods distinguish lift from prediction. Second, **regulatory and supervisory sources** support a bank-native decomposition into balances, PPNR, PD, LGD, EAD, losses, scenarios, and controls. Third, **internal bank data** must identify proprietary quantities that no public paper can supply: activation rates, wallet-share capture, cannibalization, reward response, payment response, loss elasticity, cost allocation, and terminal value. Fourth, **design conventions** make the component auditable: valuation semantics identifiers, request lineage, score-status handling, deterministic replay, and use-case approval.

The rest of the report is organized accordingly. Sections 5 through 11 preserve the literature survey and modeling framework because that evidence is the substantive basis for the component. Sections A through B translate the evidence into the actual replacement component specification. Section C records source-support limits and claim use.

For readers using the document operationally, the preserved evidence body and the new operating specification have different jobs. The evidence body explains why the component needs state-dependent spend, separate activation and use, cannibalization, lifetime states, PD/LGD/EAD, PPNR, and policy-dependent value. The operating specification explains what the replacement component must expose to other systems: decision context, valuation semantics, request and response contracts, status handling, operating modes, migration gates, and governance approval. The tables in the operating specification are therefore contract summaries; the prose around them is the primary reader path.

Evidence-to-design spine

To avoid a disjoint “survey then specification” structure, each major evidence block should be read as answering five implementation questions. First, what behavioral or financial mechanism does the literature support? Second, what component requirement follows from that mechanism? Third, what internal bank data or experiment is required before the mechanism can be parameterized for this issuer? Fourth, which output field or decomposition item changes because of the mechanism? Fifth, what validation or monitoring test would veto use of that component in production? This spine is summarized as:

evidence $\rightarrow$ model requirement $\rightarrow$ bank data $\rightarrow$ NPV output $\rightarrow$ validation gate.

For example, unemployment and liquidity evidence do not merely justify adding “macro variables” to a score. They require state-dependent purchase, payment, balance, attrition, and loss paths, with scenario tags in the output and macro-stress validation before use. Payment-choice evidence does not merely say that a customer may open a card. It requires separate issuance, activation, first-use, repeated-use, and wallet-share states, with new-card spend kept separate from all-bank incremental spend. Credit-risk and supervisory sources do not by themselves define campaign NPV, but they require the valuation component to expose PD, LGD, EAD, balances, PPNR, expected loss, funding, and capital in a form that finance, risk, and model-risk teams can reconcile. The rest of the document follows this pattern.

3 Introduction

A large U.S. retail bank considering a credit card campaign faces two related business cases. First, the bank may market a card to a prospect who is not yet a card customer. Second, it may offer a

new card to an existing client, including an existing bank customer with deposits or loans, or an existing cardholder who could add another card. In both cases, the operational decision is often framed as a propensity problem: who will respond, who will be approved, who will activate, and who will spend. That framing is useful for funnel management, but it is incomplete for capital allocation. The economically relevant object is a profit or value objective, not a response score alone. This is consistent with the direct-marketing literature on optimal selection, which ranks customers by expected net return rather than by response probability alone [16], and with the customer lifetime value (CLV) and customer-equity literatures, which define customer value as a discounted stream of expected future contribution [8, 45, 53]. The credit-card setting adds bank-specific components—losses, capital, funding, fraud, rewards, and cannibalization—to this familiar marketing objective.

Let i index customers, t monthly time, and $a \in \mathcal{A}$ a marketing or product action, such as a direct-mail offer, digital preapproval, branch offer, balance-transfer campaign, rewards bonus, or product upgrade. In the direct-mail formulation, the bank would mail customer i when expected margin times response probability exceeds contact cost:

$$d_i^* = \mathbf{1}\{p_i m_i - c_i > 0\}, \quad (2)$$

where p_i is response probability, m_i is conditional gross margin, and c_i is contact cost. In a CLV formulation, customer value is instead a discounted contribution stream:

$$\text{CLV}_i = \mathbb{E} \left[\sum_{t=0}^T \delta_t \{R_{it} - C_{it}\} \middle| \mathcal{I}_{i0} \right]. \quad (3)$$

The card-acquisition problem combines these ideas with treatment effects:

$$\begin{aligned} \Delta CF_{i,t+h}(a, \pi; s) &= \Delta PPNR_{i,t+h}(a, \pi; s) - \Delta EL_{i,t+h}(a, \pi; s) - \Delta Kchg_{i,t+h}(a, \pi; s) \\ &\quad - \Delta Tax_{i,t+h}(a, \pi; s) + \Delta RelValue_{i,t+h}(a, \pi; s), \end{aligned} \quad (4)$$

$$\Delta NPV_i(a; d, s, \pi) = -C_i^{\text{acq}}(a) + \mathbb{E} \left[\sum_{h=0}^H \delta_h \Delta CF_{i,t+h}(a, \pi; s) + \delta_H \Delta TV_{i,t+H}(a, \pi; s) \middle| \mathcal{I}_{id} \right]. \quad (5)$$

Here ΔCF is the one-period incremental cash-flow primitive, $\Delta PPNR$ is pre-provision net revenue, ΔEL is expected credit loss net of recovery, $\Delta Kchg$ is a priced capital or balance-sheet charge, ΔTax is the tax effect under the declared valuation convention, $\Delta RelValue$ is approved relationship-value spillover, C_i^{acq} is one-time acquisition or setup cost, and ΔTV is terminal value. The increments are measured against the decision-specific baseline action a_0 , under scenario and valuation bundle s , downstream policy π , and the information set $\mathcal{I}_{id}$ available at decision context d .

Thus, Equation (5) is not claimed to be a single canonical equation from one credit-card paper. It is the synthesis needed for a bank: Equation (2) contributes the expected-profit targeting logic, Equation (3) contributes the discounted lifetime stream, and causal/uplift methods contribute the requirement that all terms be incremental to the no-offer counterfactual.

Four Layers of the Valuation Problem

The proposal uses four distinct layers. The *identity layer* defines cash-flow and stock-flow accounting objects such as Equation (4); it must reconcile to finance and risk definitions. The *causal layer* estimates the effect of actions on observable short- and medium-horizon outcomes, such as response, activation, spend source, balance build, payment behavior, attrition, and loss. The *forecast layer*

projects state transitions and cash-flow drivers under a declared scenario and current state. The *policy-value layer* evaluates the discounted path under downstream account-management policy π , including future line, repricing, retention, collections, closure, hardship, and conversion rules. A module may be predictive without being causal, or causal over a short horizon without by itself identifying long-horizon policy value; the component contract should say which layer each output supports.

This counterfactual is crucial. For an existing client, observed post-campaign spend on a new card can be high even when incremental NPV is low: the card may shift spend from the bank’s old card, pull forward spend through a costly promotion, or create balances whose losses exceed finance charges. Conversely, a moderate activation probability can be valuable if it creates durable incremental purchase volume, profitable revolving balances, and low servicing and loss costs. The bank’s literature problem is therefore not a search for a single “credit-card NPV paper.” It is a synthesis problem across household finance, payment choice, credit-card economics, causal marketing, and customer lifetime value.

This paper is organized around three questions. Section 5 reviews evidence on how market conditions, employment, income, and liquidity alter spending and card economics. Section 6 reviews adoption, activation, rewards, and use of new payment instruments. Section 7 reviews customer lifetime value and duration models for lifetime expenditure. Section 8 proposes a practical model architecture for a bank. Section 9 develops a component decomposition into losses, balances, PPNR, and portfolio-management overlays. Section 10 addresses measurement and decision pitfalls that arise when NPV is used across the card funnel. Section 11 discusses empirical design and validation. Section 12 develops the endogeneity, selection, and data strategy required to make the component defensible. Section C records the source-support limits of this first-pass survey.

4 Integrated Design of the Replacement NPV Component

This section states the proposal in build terms before the detailed evidence sections. The replacement is a valuation component that converts a declared decision context, a candidate action, a customer state, product terms, scenario assumptions, and downstream policy assumptions into a decomposed incremental NPV object. The component must be modular because card value is not generated by one behavior. It is generated by a linked path of adoption, spend, payments, balances, losses, rewards, funding, capital, servicing, attrition, and relationship effects.

For every module, the design rule is the same. The public literature or supervisory source identifies a mechanism and a modeling slot. The bank’s internal data or experiment estimates the issuer-specific parameter. The component exposes the resulting path or cash-flow driver in the NPV response. Validation determines whether that module may be used for a given decision context. This is the practical meaning of evidence integration in this proposal.

4.1 Module 1: spend, macro, and employment state

The spend module forecasts purchase volume, transaction count, ticket size, and merchant-category mix under the declared action and macro scenario. The literature on unemployment, income shocks, liquidity, and macro disruptions supports the mechanism that spending is state dependent. The component therefore should not use a static average spend curve for all campaign conditions. It should condition on customer liquidity, payroll or deposit signals where permitted, utilization, local labor-market variables, rates, inflation, and scenario tags.

The bank must estimate how those variables map into its own card spend, payment rate, balance build, and attrition. Public papers can justify including unemployment or liquidity states; they

cannot identify this bank’s category spend elasticity or routing share to the new card. The output implication is that NPV responses should expose spend and interchange by category and horizon, plus scenario sensitivity. The validation implication is that macro backtests, time-based holdouts, and segment calibration can veto the spend module even if average first-spend prediction is strong.

4.2 Module 2: new-card adoption, activation, use, and cannibalization

The adoption and usage module separates offer response, application or acceptance, approval where relevant, booking, activation, first use, repeated use, wallet-share capture, dormancy, attrition, and charge-off. Payment-choice and rewards evidence supports the distinction between having a payment instrument and using it. It also supports modeling rewards and promotions as treatments that can change activation, spend, debt, and reactivation.

For existing clients, the core output is not raw new-card spend. The component must distinguish new-card spend, all-bank incremental card spend, debit or deposit substitution where observed, old-bank-card cannibalization, and uncertain outside-card or new-consumption components. The bank must estimate these quantities using randomized holdouts, campaign experiments, or credible quasi-experimental designs. The response object should label activation, first-use probability, repeated-use probability, new-card spend lift, all-bank-spend lift, reward cost, balance lift, loss lift, and cannibalization. A model that predicts new-card spend but cannot identify incremental all-bank value should remain diagnostic for cross-sell decisions.

For prospect acquisition and existing-client cross-sell, the module should also estimate an outside-wallet state and an offer-specific movable-wallet state. Outside wallet is the customer’s spend or revolving balance not currently on the bank’s card products. Movable wallet is the fraction of that outside wallet that the specific offer can plausibly move to the bank over the declared horizon. It should depend on product terms, rewards, promotional duration, channel, card-on-file friction, relationship depth, segment, macro state, and time since offer. The component should expose whether outside-wallet and movable-wallet quantities are directly observed, internally anchored, externally benchmarked, experimentally identified, quasi-experimental, model-imputed, or scenario-only.

4.3 Module 3: lifetime state, attrition, dormancy, and terminal value

The lifetime module turns early behavior into a state path. A card customer can be a transactor, revolver, promotional borrower, cash-advance user, dormant account, retained account, converted account, delinquent account, or charged-off account. CLV, customer-base, and duration models support the use of transaction frequency, recency, monetary value, active-state inference, and relationship duration, but card NPV must extend those models to balances, losses, rewards, funding, and servicing.

The component should estimate transition probabilities among these states and carry state-specific terminal-value assumptions. Terminal value should not be one hidden scalar multiple; it should depend on state, vintage, months on book, product, relationship depth, risk, macro state, and downstream policy. The bank must validate lifetime state forecasts with vintage holdouts, dormancy calibration, attrition calibration, and terminal-value sensitivity. If terminal value dominates rankings, the response should expose that fact as a sensitivity or warning.

4.4 Module 4: balances, payments, lines, promotions, and cash advances

The balance module enforces stock-flow consistency. Purchases, cash advances, balance transfers, fees, interest, payments, credits, charge-offs, and adjustments should reconcile to statement and

principal balances. Credit line, APR, promo duration, balance-transfer terms, rewards, and disclosures are candidate action terms, not passive descriptors. They can change both revenue and risk through utilization, revolve behavior, payment rates, and EAD.

The bank must estimate line and term response from its own account history, experiments, policy changes, and approved quasi-experimental variation. The component output should report line, utilization, purchase balances, cash-advance balances, balance-transfer balances, promotional balances, payments, finance charges, fees, funding cost, and balance-linked EAD. A balance forecast that violates stock-flow accounting or mixes principal and non-principal balances should fail component validation.

4.5 Module 5: losses, PD, LGD, EAD, vintage, and balance subledgers

The loss module uses the bank-standard PD/LGD/EAD vocabulary, but it must be connected to the card lifecycle. PD should be a transition or hazard over delinquency and charge-off states. LGD should reflect recovery, collections, bankruptcy, hardship programs, and macro state. EAD should be linked to balances, line, utilization, cash advances, balance transfers, accrued fees, interest, and undrawn draw behavior.

The component should expose expected loss and its drivers by horizon and option. It should also preserve vintage, months-on-book, and principal/non-principal balance dimensions, because newly booked accounts season differently and promotions can shift both timing and composition of loss. Public credit-risk literature supports the structure, but the bank must validate calibration and treatment response using internal vintages, booked account outcomes, line-policy tests, and stress periods. A line or promo option should not be ranked on revenue unless the linked EAD, PD, LGD, capital, and downside risk paths are also valid.

4.6 Module 6: PPNR, rewards, funding, capital, tax, cost, and relationship value

The finance module converts behavior into cash flows. It should decompose incremental value into net interest income, non-interest income, non-interest expense, expected loss, capital cost, tax treatment, acquisition cost, terminal value, and approved relationship value. PPNR and stress-testing sources support separating balances, revenue, expense, and losses under scenarios; CLV literature supports discounting future contribution; bank governance requires the decomposition to reconcile to finance and risk definitions.

Rewards, signup bonuses, servicing, fraud, disputes, collections, charge-off sale economics, funding, liquidity, capital, and tax must not be afterthoughts. They are often what turns high spend into low or negative value. Deposit or other-product value should be included only when the card action is credibly causal for relationship change; otherwise it should be excluded or reported separately. The component should expose both the decomposed contribution path and the valuation-semantics identifier that defines discounting, funding, capital, tax, cost allocation, terminal value, and relationship-value rules.

4.7 Module 7: sequential policy dependence

The component must tag the downstream policy assumed when valuing an action. Acquisition value depends on future approval, line, repricing, retention, collections, closure, and conversion policies. Underwriting and line choices change the population and state distribution that account-management systems later see. Account-management changes can invalidate upstream acquisition NPV if the old model assumed different future treatment rules.

The practical requirement is a downstream-policy assumption bundle. Each NPV response should state which policy bundle was used, and each production use case should be approved for that bundle. If the bank changes line-management, collections, retention, closure, or conversion policy, upstream NPV should be revalidated. This keeps the component from optimizing a future operating world the bank no longer runs.

4.8 How the modules connect to caller systems

Caller systems do not need to know every internal model detail, but they must receive a value object whose meaning is safe for their decision. A marketing caller needs contact-versus-suppress value, campaign cost, response and activation lift, cannibalization, expected loss, and portfolio constraints. An underwriting caller needs approve-versus-decline or approve-with-terms value under credit policy, with loss, capital, compliance, and adverse-action boundaries handled by the caller. A line-assignment caller needs option-level value, usage, utilization, EAD, loss, capital, and status for each feasible line. An existing-account treatment caller needs forward value under retention, conversion, promo, closure, or no-treatment alternatives. Finance and risk consumers need decompositions for reconciliation, not authority to treat the component as the official ledger, CECL, or stress platform.

This is why the operating contracts later in the document exist. They are not the product. They are the guardrails that keep a sophisticated but fragile NPV estimate from becoming a misleading universal score.

5 Market Conditions, Employment, and Propensity to Spend

5.1 Income, unemployment, and liquidity

The first component of credit-card NPV is the customer’s capacity and propensity to spend. The relevant empirical papers do not estimate the bank’s Equation (5) directly. They estimate causal or quasi-causal effects of income, employment, and liquidity variables on spending or debt. Their contribution to our model is a set of state variables and elasticities for the spend, balance, and risk modules.

Ganong and Noel [25] use de-identified bank account data to study spending around unemployment and the predictable income drop at unemployment insurance exhaustion. A stylized event-study version of the object is

$$y_{i\tau} = \alpha_i + \lambda_\tau + \sum_{k \neq -1} \beta_k \mathbf{1}\{\tau_i = k\} + \varepsilon_{i\tau}, \quad (6)$$

where $y_{i\tau}$ is spending for household i at event time τ relative to an unemployment or benefit-exhaustion event, α_i absorbs household fixed effects, and the β_k trace the spending path around the event. The important modeling implication is not merely “unemployment matters.” It is that the card spend forecast should include event-time variables such as job loss, payroll interruption, benefit receipt, and benefit exhaustion, because spend can move non-smoothly around predictable income events.

Johnson et al. [32] study the 2001 tax rebates using variation in rebate timing. A stylized specification is

$$\Delta c_{it} = \alpha_i + \lambda_t + \gamma \text{Rebate}_{it} + \eta' H_i \text{Rebate}_{it} + u_{it}, \quad (7)$$

where γ is a marginal propensity to consume out of the transfer and H_i contains heterogeneity variables such as income or liquid-wealth status. For a credit-card model, this says that transfers,

payroll changes, and liquid wealth proxies should enter monthly purchase-volume and payment-rate equations, with heterogeneous effects. It does not identify the share of extra consumption that will route through our card.

Gross and Souleles [26] provide the closest direct credit-card evidence. They study account-level card and bureau data and estimate how debt responds to credit limits and interest rates. A stylized account-level equation is

$$\Delta B_{it} = \alpha_i + \lambda_t + \theta \Delta \text{Limit}_{it} + \rho \Delta \text{APR}_{it} + \psi' X_{it} + e_{it}, \quad (8)$$

with stronger responses among initially high-utilization or liquidity-constrained borrowers. This should enter our future model through the balance and payment modules: line assignment and APR are not passive controls; they are treatment components that affect revolving balances, utilization, finance charges, and expected loss.

5.2 Rates, regulation, and product terms

Credit-card NPV is also shaped by prices, fees, disclosures, repayment rules, and regulatory constraints. Agarwal et al. [2] study the Credit Card Accountability Responsibility and Disclosure Act using a large account-level card panel. The paper is not a spend-propensity model, and this survey should not use it as if it supplied a universal activation or balance elasticity. Its modeling value is different: it shows that product terms are governed economic objects. Fees, disclosures, and repayment nudges can change both issuer revenue components and consumer repayment behavior, so the NPV model cannot treat APRs, fees, and promotional terms as free scalar revenue add-ons.

Formally, the campaign action should be represented as a vector of product terms, not as a binary contact flag:

$$a_{it} = \left(\ell_{it}, r_{it}^{\text{purchase}}, r_{it}^{\text{cash}}, r_{it}^{\text{BT}}, r_{it}^{\text{promo}}, T_{it}^{\text{promo}}, f_{it}, q_{it}, b_{it}, d_{it} \right) \in \mathcal{A}_{it}(X_{it}, M_{gt}, G_t). \quad (9)$$

Here ℓ is the credit line; the r terms are purchase, cash-advance, balance-transfer, and promotional APRs; T^{promo} is promotional duration; f is the vector of annual, late, balance-transfer, cash-advance, foreign-transaction, and waiver terms; q is the reward earn or redemption schedule; b is a signup or statement-credit bonus; and d is the disclosure, minimum-payment, autopay, and digital-wallet provisioning design. The feasible set $\mathcal{A}_{it}$ is indexed by customer state, macro state, and governance state G_t because underwriting policy, compliance rules, disclosure requirements, risk appetite, pricing floors, and operational constraints limit which offers can be made.

This notation connects the CARD Act evidence to the later NPV architecture in two ways. First, regulation changes the feasible set and sometimes the cash-flow mapping:

$$a_{it} \in \mathcal{A}_{it}(G_t), \quad (10)$$

$$CF_{i,t+1} = c(Z_{it}, X_{it}, M_{gt}, a_{it}; G_t). \quad (11)$$

A rule change, fee cap, disclosure change, or repayment-allocation rule is therefore not merely an external dummy variable. It changes either the offers available to the bank or the function that maps behavior into interest, fees, payments, and balances. Second, product terms are treatments that enter the behavioral transition system. A reduced-form version is

$$\Pr(Z_{i,t+1} = z' \mid Z_{it}, X_{it}, M_{gt}, a_{it}, G_t), \quad (12)$$

$$\mathbb{E}[Y_{i,t+1} \mid Z_{i,t+1}, X_{it}, M_{gt}, a_{it}, G_t], \quad Y \in \{S, B, \text{Pay}, D, Q\}. \quad (13)$$

Thus APR, line, fees, rewards, and disclosures can affect activation, purchase volume, revolving balances, payments, delinquency, rewards liability, servicing cost, and attrition. They should not be appended only after a response model has already selected customers.

The mathematical bridge to Gross and Souleles [26] is direct. Equation (8) says that limit and APR changes can move card debt, especially for liquidity-constrained accounts. In the NPV engine, the same terms must feed both revenue and risk:

$$B_{i,t+1} = g_B(B_{it}, Pay_{it}, S_{it}, \ell_{it}, r_{it}, f_{it}, X_{it}, M_{gt}) + \varepsilon_{i,t+1}^B, \quad (14)$$

$$Int_{i,t+1} = r_{it}^{\text{revolve}} B_{i,t+1}^{\text{revolve}} + r_{it}^{\text{cash}} B_{i,t+1}^{\text{cash}} + r_{it}^{\text{BT}} B_{i,t+1}^{\text{BT}}, \quad (15)$$

$$EL_{i,t+1} = PD_{i,t+1}(Z_{it}, Util_{it}, X_{it}, M_{gt}, a_{it}) LGD_{i,t+1} EAD_{i,t+1}(B_{i,t+1}, \ell_{it}, a_{it}). \quad (16)$$

Raising APR can mechanically increase the rate applied to revolving balances, but it can also change borrowing, payment, attrition, and default behavior. Increasing the line can raise usage and finance revenue, but it can also raise utilization dynamics, exposure at default, capital, and downside risk. The correct object for decisioning is therefore a total derivative of value with respect to the governed offer vector, not a partial revenue calculation holding behavior fixed. Let $\mu_Y(a) = \mathbb{E}[Y_i(a) \mid \mathcal{I}_{it}]$ for $Y \in \{Z, S, B, Pay, D, Q\}$. For any feasible local perturbation v of the offer vector,

$$\begin{aligned} D_v \mathbb{E}[\Delta NPV_i(a)] &= \nabla_a \mathbb{E}[\Delta NPV_i(a, \mu)]^\top v \Big|_{\mu=\mu(a)} \\ &+ \sum_{Y \in \{Z, S, B, Pay, D, Q\}} \frac{\partial \mathbb{E}[\Delta NPV_i(a, \mu)]}{\partial \mu_Y} \Big|_{\mu=\mu(a)} D_v \mu_Y(a). \end{aligned} \quad (17)$$

The first term is the mechanical cash-flow effect of the term sheet; the second term is the behavioral response channel. The second term is precisely why the rates and regulation literature belongs before the usage, CLV, loss, balance, and PPNR sections.

This also clarifies how product terms should be validated. The promotion criterion is not whether a high-APR or high-fee offer raises modeled revenue conditional on the same balance path. It is whether the feasible offer $a \in \mathcal{A}_{it}(G_t)$ increases counterfactual value after behavioral response, regulation, cannibalization, expected loss, capital, funding, rewards, and servicing:

$$\Delta NPV_i(a; G_t) = -\Delta C_i^{\text{acq}}(a) + \sum_{h=0}^H \delta_h \mathbb{E}[\Delta CF_{i,t+h}^{\text{terms}}(a; G_t) \mid \mathcal{I}_{it}], \quad (18)$$

$$\Delta CF_{i,t+h}^{\text{terms}}(a; G_t) = \Delta PPNR_{i,t+h}(a; G_t) - \Delta EL_{i,t+h}(a; G_t) - \Delta K_{i,t+h}(a; G_t) + \Delta RelValue_{i,t+h}(a). \quad (19)$$

Equation (18) is the handoff to the component decomposition in Section 9: promotional APRs and balance-transfer terms become promo states and balance subledgers; fees and waivers become non-interest income; rewards become expense and liability; APR/line choices affect balances, EAD, and capital; disclosure and repayment design enter payment and attrition transitions; and all terms remain subject to the feasible governed action set.

5.3 Macro shocks and category substitution

Market shocks can change not only the level of spending but also the composition of card spending. Baker et al. [6] use transaction-level household financial data to study consumption during the COVID-19 shock. Their evidence documents large changes in spending levels and categories during

Table 1: How Section 5 evidence should enter a bank model.

Literature	Mathematical object	Model use
Ganong and Noel [25]	Event-time spending path around unemployment and benefit exhaustion.	Add job-loss, payroll interruption, benefit receipt, and exhaustion indicators to spend, payment, attrition, and loss modules.
Johnson et al. [32]	Marginal propensity to consume out of transfer timing, with heterogeneity.	Add income/transfer shocks and liquidity interactions to spend and payment equations; do not assume routing to our card.
Gross and Souleles [26]	Account-level response of card debt to credit limits and APR.	Treat line and APR as treatment components affecting balances, finance revenue, utilization, and loss.
Agarwal et al. [2]	Regulation-induced changes in fees, disclosures, repayment behavior, and issuer revenue components.	Treat APRs, fees, rewards, disclosures, and repayment rules as governed terms $a_{it} \in \mathcal{A}_{it}(G_t)$ that feed transition, balance, loss, PPNR, and promo-state equations rather than unconstrained revenue levers.
Baker et al. [6]	Transaction-level response of spending levels and categories to a macro shock.	Forecast category-level spend and interchange/rewards/fraud mix under macro scenarios.

the epidemic. For card NPV, category shifts matter because merchant-category mix affects interchange, rewards expense, fraud risk, dispute rates, and charge-off correlation. A recession, health shock, inflation shock, or local labor-market shock can lower discretionary purchase volume while increasing borrowing demand and credit risk.

The practical lesson is that spend models should be time varying and category-aware. A customer with high pre-campaign grocery and travel spend is not equally valuable in every macro state. A useful reduced-form component is

$$\log(1 + S_{ij,t+1}) = \mu_i + \lambda_t + \kappa_j + \beta'_j M_{gt} + \chi'_j X_{it} + \omega'_j (M_{gt} \times X_{it}) + \epsilon_{ij,t+1}, \quad (20)$$

where $S_{ij,t+1}$ is spend in merchant category j , M_{gt} is a local or national macro vector for geography g , and X_{it} includes customer liquidity, payroll, utilization, and relationship variables. Equation (20) is not lifted from one paper; it is the model slot suggested by the income, liquidity, and macro-shock evidence.

5.4 Implication for the NPV component

The macro and employment evidence should be incorporated as a dynamic state requirement, not as a cosmetic scenario overlay. The replacement component should carry macro, local labor-market, rate, inflation, payroll, benefit, and liquidity variables into the spend, payment, balance, attrition, and loss modules. It should also return the scenario identifier and sensitivity of major NPV components to macro assumptions. For marketing and cross-sell, this means the model must distinguish customers whose expected value comes from durable purchase volume from customers whose value

depends on fragile revolve or liquidity stress. For line and offer decisions, it means the line, APR, and promo terms must be modeled as actions that can change both revenue and EAD, not as exogenous descriptors.

The public literature does not supply this bank’s elasticity of card spend, payment rate, or default to unemployment and liquidity shocks. Those parameters require issuer transaction data, bureau refreshes, payroll or deposit signals where permitted, campaign holdouts, and vintage backtests. The output implication is concrete: the response object should expose category spend, balances, payments, utilization, PD, LGD, EAD, expected loss, and PPNR by scenario and horizon. The validation implication is equally concrete: macro-stress backtests, local labor-market drift monitoring, and segment-level calibration can veto use of a model even when average response or spend prediction is strong.

6 Will the Newly Issued Card Be Used?

6.1 Adoption is not usage

The second requested problem is the probability that a customer who receives a new card will use that card. The payment-choice literature warns against collapsing this into a single response probability. Koulayev et al. [33] estimate a structural model of adoption and use of payment instruments by U.S. consumers. Their central relevance here is that possession of a payment instrument and use of that instrument are distinct choices. A customer may open a card, activate it, and still use another issuer’s card, debit card, cash, ACH, or a digital wallet at the point of sale.

Mathematically, the useful structure is a two-stage adoption/use problem. A stylized version is:

$$A_{im}^* = Z_i' \alpha_m + \nu_{im}, \quad A_{im} = \mathbf{1}\{A_{im}^* > 0\}, \quad (21)$$

$$U_{ijm} = X_{ijm}' \beta_m + \phi_m A_{im} + \varepsilon_{ijm}, \quad m_{ij}^* = \arg \max_{m \in \mathcal{M}_i} U_{ijm}. \quad (22)$$

Here A_{im} denotes whether consumer i has adopted payment instrument m , U_{ijm} is the utility of using instrument m for transaction context j , and $\mathcal{M}_i$ is the set of adopted instruments available to consumer i . The exact specification in Koulayev et al. [33] is richer, but Equations (21)–(22) capture the modeling lesson for us: “has the card” gates the choice set, while “uses the card” is a conditional payment-choice decision. Our card model should therefore estimate activation and usage conditional on the customer’s competing payment portfolio rather than treating issuance as usage.

For a credit-card campaign, the operational funnel should therefore separate:

1. offer exposure and response;
2. application and approval;
3. card opening and physical or digital activation;
4. first purchase;
5. repeated purchase;
6. share-of-wallet capture;
7. durable active state, dormancy, closure, or charge-off.

Each transition has different covariates and different economic meaning. Approval is partly a risk and underwriting outcome. Activation is partly a friction and onboarding outcome. First purchase is an early usage outcome. Share-of-wallet is a competitive payment-routing outcome. Durable activity is a lifetime-value outcome.

6.2 Rewards, reactivation, and early spend

Rewards can alter usage, but the bank must distinguish profitable use from costly promotional pull-forward. Agarwal et al. [1] study cash-back rewards using administrative data from a large U.S. financial institution. The important point for our later model is that the paper is not just a qualitative statement that “rewards matter.” It suggests three account-level response objects:

$$A_{it}^{\text{react}} = \mathbf{1}\{S_{i,t-3:t-1} = 0, S_{it} \geq s_0\}, \quad (23)$$

$$S_{it} = \text{monthly purchase spend on the card}, \quad (24)$$

$$B_{it} = \text{monthly card debt or revolving balance}. \quad (25)$$

Here A_{it}^{react} is a reactivation indicator among accounts that were inactive before the reward, S_{it} is the usage outcome, and B_{it} is the balance outcome. These are exactly the three objects a bank needs for a new card: did the account become active, how much spend did it generate, and did it also create debt exposure?

A reduced-form implementation for a randomized or quasi-randomized reward offer can be written as

$$Y_{it} = \alpha_i + \lambda_t + \tau \text{Reward}_{it} + \eta' X_{it} + \epsilon_{it}, \quad (26)$$

where Y_{it} may be purchase spend, revolving debt, or an indicator for reactivation among previously inactive cardholders. The estimand is

$$\tau_Y = \mathbb{E}[Y_{it}(r = 1) - Y_{it}(r = 0)], \quad (27)$$

with r denoting the reward treatment. In a bank implementation this should be estimated heterogeneously,

$$\tau_Y(x, r) = \mathbb{E}[Y_i(r) - Y_i(0) \mid X_i = x], \quad (28)$$

where X_i includes prior inactivity, existing card holdings, pre-period merchant mix, line utilization, relationship depth, channel, credit quality, and macro state. Equation (28) is the bridge from Agarwal et al. [1] to the later model: it gives a segment-level reward response surface rather than a single portfolio average.

The output of this block should feed the NPV simulator in four places:

$$\Pr(Z_{i,t+1} = \text{active} \mid Z_{it}, X_{it}, a) \leftarrow \text{activation/reactivation effect}, \quad (29)$$

$$\mathbb{E}[S_{i,t+1} \mid Z_{i,t+1}, X_{it}, a] \leftarrow \text{spend effect}, \quad (30)$$

$$\mathbb{E}[B_{i,t+1} \mid Z_{i,t+1}, X_{it}, a] \leftarrow \text{debt/balance effect}, \quad (31)$$

$$C_{i,t+1}^{\text{reward}}(a) = q_a S_{i,t+1} + F_a + R_a^{\text{signup}}, \quad (32)$$

where q_a is the earn rate or reward cost per dollar, F_a is fixed program cost, and R_a^{signup} is signup or statement-credit cost. The promotion is attractive only if the incremental contribution implied by (29)–(32) is positive after cannibalization and losses. This is the concrete way to avoid importing a headline reward effect as if it were a universal activation parameter.

Behavioral evidence also suggests that payment instruments can affect purchase behavior. Prelec and Simester [41] find in experimental settings that willingness to pay can be higher under credit-card payment than cash payment. For a bank NPV model, this is mechanism evidence: payment method may change the purchase decision, not simply route a fixed purchase through a different rail. The result is not a portfolio-level NPV estimate and should not be used as a direct calibration without issuer data.

The conservative model use is a mechanism prior for the ticket-size component. Let N_{ijt} be transaction count in category j and $\bar{s}_{ijt}$ average ticket size. Then

$$S_{ijt} = N_{ijt}\bar{s}_{ijt}, \quad (33)$$

$$\log \bar{s}_{ij,t+1} = \mu_i + \kappa_j + \beta' X_{it} + \theta_1 \text{CreditCard}_{ij,t+1} + \theta_2 \text{RewardSalience}_{ij,t+1} + \epsilon_{ij,t+1}. \quad (34)$$

The Prelec and Simester [41] result supports allowing θ_1 or θ_2 to differ from zero, but the bank must estimate those parameters with transaction data. This block is explanatory unless validated on issuer data; it should not be a hard-coded multiplier in NPV.

6.3 Cannibalization and incremental usage

For existing clients, the central challenge is cannibalization. A newly issued card can create genuinely incremental spend, shift spend from another issuer, shift spend from the bank's existing card, move checking/debit activity onto credit, or induce balances whose loss and reward costs exceed revenue. To make this estimable, define the following observed spend aggregates over horizon H :

$$S_{iH}^{\text{new}} = \text{spend on the newly issued card}, \quad (35)$$

$$S_{iH}^{\text{oldbank}} = \text{spend on the bank's pre-existing cards}, \quad (36)$$

$$S_{iH}^{\text{debit}} = \text{bank-observed debit/checking purchase activity}, \quad (37)$$

$$S_{iH}^{\text{bank}} = S_{iH}^{\text{new}} + S_{iH}^{\text{oldbank}}. \quad (38)$$

When outside-card spend is not observed, competitor-card substitution is latent and must be inferred from bureau/payment-network aggregates, wallet estimates, or experimental lift in total bank spend. The observed new-card spend path should therefore be decomposed before it is valued:

$$S_{iH}^{\text{new}}(a) = S_{iH}^{\text{new cons}}(a) + S_{iH}^{\text{comp shift}}(a) + S_{iH}^{\text{oldbank shift}}(a) + S_{iH}^{\text{debit shift}}(a) + S_{iH}^{\text{promo timing}}(a) + e_{iH}^S(a). \quad (39)$$

Equation (39) is an accounting discipline for model design, not a claim that every term is directly observed. The first term is genuinely new consumption. The second is spend shifted from competitor cards. The third is internal bank-card cannibalization. The fourth is debit or deposit-rail substitution. The fifth is intertemporal pull-forward or decay created by promotions. The last term collects measurement error and unobserved routing changes. These terms have different economics. Competitor shift can be valuable to the issuer; old-bank-card shift may merely move interchange and reward cost across products; debit substitution may reduce deposit economics; promotion timing may disappear after the offer; and all terms must still be netted against rewards, fraud, funding, servicing, losses, and capital.

$$W_{iH}^{\text{out}} = S_{iH}^{\text{comp}} + B_{iH}^{\text{comp,rev}} + U_{iH}^{\text{outside}}, \quad W_{iH}^{\text{move}}(a) = m_{iH}(a) W_{iH}^{\text{out}}. \quad (40)$$

Here W_{iH}^{out} is an outside card-wallet state combining competitor-card spend, competitor revolving balances, and other outside payment or borrowing components relevant to the decision; $m_{iH}(a)$ is the offer-specific movable fraction; and $W_{iH}^{\text{move}}(a)$ is the portion the offer can plausibly move to the

bank. The model should distinguish this structural movable fraction from an operational reporting ratio such as observed new-card spend divided by estimated outside wallet. A large outside wallet does not imply large movable wallet, and large movable wallet does not imply positive NPV after rewards, losses, and funding.

$$\begin{aligned}\Delta S_{iH}^{\text{new}}(a) = & \Delta S_{iH}^{\text{new cons}}(a) + \Delta S_{iH}^{\text{comp shift}}(a) + \Delta S_{iH}^{\text{oldbank shift}}(a) \\ & + \Delta S_{iH}^{\text{debit shift}}(a) + \Delta S_{iH}^{\text{promo timing}}(a) + \Delta e_{iH}^S(a).\end{aligned}\quad (41)$$

Equation (41) is the version that belongs in incremental NPV. It says that the treatment effect on new-card spend should be allocated to sources before it is valued. A response model that estimates only ΔS^{new} can rank new-card volume, but it cannot by itself rank incremental NPV. The estimands are:

$$\tau_i^{\text{new card spend}}(a) = \mathbb{E}[S_{i,t:t+H}^{\text{new}}(a) - S_{i,t:t+H}^{\text{new}}(0) \mid \mathcal{I}_{it}], \quad (42)$$

$$\tau_i^{\text{bank spend}}(a) = \mathbb{E}[S_{i,t:t+H}^{\text{all bank cards}}(a) - S_{i,t:t+H}^{\text{all bank cards}}(0) \mid \mathcal{I}_{it}], \quad (43)$$

$$\text{Cannibalization}_i(a) = \tau_i^{\text{new card spend}}(a) - \tau_i^{\text{bank spend}}(a). \quad (44)$$

Equation (42) can be positive while Equation (43) is small or even negative. That is why new-card activation and new-card spend are not sufficient promotion criteria.

In a randomized campaign, these are directly estimable by treatment-control differences:

$$\hat{\tau}^{\text{new}}(x) = \mathbb{E}[S_{iH}^{\text{new}} \mid T_i = 1, X_i = x] - \mathbb{E}[S_{iH}^{\text{new}} \mid T_i = 0, X_i = x], \quad (45)$$

$$\hat{\tau}^{\text{bank}}(x) = \mathbb{E}[S_{iH}^{\text{bank}} \mid T_i = 1, X_i = x] - \mathbb{E}[S_{iH}^{\text{bank}} \mid T_i = 0, X_i = x], \quad (46)$$

$$\hat{\kappa}(x) = 1 - \frac{\hat{\tau}^{\text{bank}}(x)}{\hat{\tau}^{\text{new}}(x)} \quad \text{when } \hat{\tau}^{\text{new}}(x) > 0. \quad (47)$$

Here $\hat{\kappa}(x)$ is the estimated cannibalization rate from the bank's own existing-card book. If $\hat{\kappa}(x)$ is close to one, most new-card spend is shifted from old bank cards; if it is close to zero, new-card spend is mostly incremental to the bank. If it is negative, the new card may also lift old bank card spend, for example through relationship deepening.

The latent competitor-transfer component can be represented as

$$\tau_i^{\text{competitor shift}}(a) = \tau_i^{\text{bank spend}}(a) - \tau_i^{\text{debit shift}}(a) - \tau_i^{\text{new consumption}}(a), \quad (48)$$

The debit-shift term can sometimes be estimated from the bank's checking and debit data. The new-consumption term is harder to identify without network or merchant-panel data. Equation (48) should therefore be treated as a decomposition with an uncertainty interval, not as a precisely observed quantity.

Each component of Equation (39) should carry an evidence grade. For example, old-bank-card shift can often be internally anchored from the bank's own card book; debit shift can be internally anchored when the customer has deposit activity at the bank; competitor shift may be experimentally identified by all-bank-spend lift in a holdout design, externally benchmarked from bureau or network aggregates where available, or model-imputed when no anchor exists; and promotion timing should be validated from post-promo decay. The component should not allow model-imputed outside wallet to be reported as experimentally identified wallet capture.

Randomized campaign holdouts are the cleanest way to separate organic propensity from treatment effect. When randomization is unavailable, the causal-inference and uplift literatures provide

tools, but they do not remove the need for credible identification. The propensity-score framework of Rosenbaum and Rubin [43], true-lift marketing models such as Lo [36], causal forests [54], and double/debiased machine learning [18] are useful methodological references. In a regulated bank environment, they should be used with overlap checks, sensitivity analysis, segment calibration, and careful model-governance review.

For later modeling, the cleanest operational implementation is to estimate a conditional average treatment effect for each downstream outcome,

$$\tau_Y(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x], \quad (49)$$

where Y may be activation, first purchase, first-90-day new-card spend, all-bank-card spend, revolving balance, loss, or full NPV. The causal/uplift module should feed the simulation engine by replacing raw predictions $\mathbb{E}[Y_i \mid T_i = 1, X_i]$ with counterfactual contrasts $\mathbb{E}[Y_i(1) - Y_i(0) \mid X_i]$. If randomization is not available, the model must record the identification assumption and cannot be promoted solely on predictive fit.

The handoff to Equation (5) is explicit. Let r_j^{int} be net interchange by merchant category, $q_j(a)$ reward cost by category and offer, m_{it}^{finance} net finance margin on balances, and ℓ_{it} expected loss. Then a horizon- H incremental contribution block is

$$\begin{aligned} \Delta\Pi_{iH}(a) &= \sum_j \{r_j^{\text{int}} - q_j(a)\} \hat{\tau}_{ijH}^{\text{bank}}(a) + \hat{\tau}_{iH}^{\text{finance}}(a) - \hat{\tau}_{iH}^{\text{loss}}(a) - C_i^{\text{campaign}}(a) - C_i^{\text{servicing}}(a), \quad (50) \\ \Delta\text{NPV}_i(a) &= \sum_{h=0}^H \delta_h \mathbb{E}[\Delta\Pi_{ih}(a) \mid \mathcal{I}_{i0}]. \quad (51) \end{aligned}$$

The key design choice is that interchange and reward revenue use $\hat{\tau}^{\text{bank}}$, not raw new-card spend S^{new} . This is how the cannibalization model changes the optimization target.

6.4 Implication for the NPV component

The adoption, rewards, and cannibalization literature implies that the component must model a funnel of states rather than a single probability that “the customer uses the card.” The minimum state sequence is offer exposure, application or acceptance, approval where relevant, booked account, activation, first purchase, repeated use, wallet-share capture, dormancy, attrition, and charge-off. Each transition has its own covariates, counterfactual, and economic meaning. A marketing caller may care about contact-versus-suppress; an underwriting caller may care about approve-versus-decline; an activation caller may care about treatment-versus-no-treatment after the account is booked. The NPV component should value those contrasts separately and should not promote a model simply because it predicts booked-account spend.

For existing clients, the response object should report at least two distinct spend concepts: new-card spend and all-bank incremental spend. The first is a card-usage diagnostic; the second is the input to bank value after cannibalization. Where the bank can observe old-card, debit, deposit, or other relationship behavior, those channels should be measured explicitly. Where outside-card behavior or new consumption is latent, the component should return an uncertainty range or warning rather than treat the decomposition as known. This bridge is what connects the literature evidence to the later request and response contracts: callers must declare the decision context, and the response must label activation, first use, new-card spend, all-bank spend, balances, reward cost, and cannibalization separately.

Table 2: How Section 6 evidence should enter a bank model.

Literature	Mathematical object	Model use
Koulayev et al. [33]	Structural adoption and conditional use of payment instruments.	Separate opened/activated state from conditional card-use choice; include competing payment instruments and transaction context.
Agarwal et al. [1]	Treatment effect of rewards on spend, debt, and reactivation.	Estimate $\tau_Y(x, r)$ for reactivation, spend, and balances; feed the activation transition, conditional spend, balance, and reward-cost blocks in Equations (29)–(32).
Prelec and Simester [41]	Experimental effect of payment medium on willingness to pay.	Use as mechanism support for the ticket-size equation (34); estimate the payment-medium coefficient on issuer transaction data before using it in NPV.
Uplift/CATE literature	$\mathbb{E}[Y(1) - Y(0) \mid X = x]$ for activation, spend, balance, loss, and NPV.	Estimate new-card lift, all-bank-card lift, and cannibalization in Equations (45)–(47); feed incremental contribution through Equation (50).

7 Lifetime Expenditure and Customer Lifetime Value

7.1 Customer-base models

The third requested problem is how to model lifetime expenditure of a credit card user. Marketing science provides a natural starting point. Schmittlein et al. [46] develop a model for inferring which customers are still active and what they will do next from transaction timing and frequency. This is directly relevant for card accounts because a customer can silently become inactive before formal closure.

Fader et al. [20] propose the BG/NBD model as a tractable alternative to the Pareto/NBD model for predicting repeat-purchase behavior. Fader et al. [21] connect recency, frequency, and monetary value to customer lifetime value. These models are useful baselines for transaction frequency, active/dormant status, and spend per transaction. They are especially relevant for transactors whose value is driven primarily by purchase volume and interchange rather than finance charges.

Relationship duration is another core object. Bolton [15] models the duration of a customer’s relationship with a continuous service provider. Credit cards have similar duration dynamics: fees, service quality, fraud events, line changes, disputes, rewards devaluations, and pricing changes can alter retention, dormancy, and usage.

7.2 Why card NPV is broader than plain CLV

Credit cards are not a pure repeat-purchase setting. A bank needs the lifetime path of at least six linked processes:

$$S_{it} = \text{purchase spend and transaction count}, \quad (52)$$

$$B_{it} = \text{statement and revolving balance}, \quad (53)$$

$$P_{it} = \text{payment amount and payment rate}, \quad (54)$$

$$D_{it} = \text{delinquency/default state}, \quad (55)$$

$$Z_{it}^{\text{use}} = \text{active, dormant, closed, or charged-off usage state}, \quad (56)$$

$$Q_{it} = \text{rewards, fraud, disputes, servicing, and other costs}. \quad (57)$$

These processes are jointly determined. High spend can be profitable when it is durable and low loss, but low value when driven by high rewards cost or low-margin categories. Revolving balances can raise interest income and expected loss simultaneously. A durable active customer may be unprofitable if servicing, fraud, or rewards costs dominate revenue.

Thus, a bank should treat CLV models as transaction and duration components of a broader card-NPV engine. Lifetime expenditure is necessary but not sufficient: the model must translate expenditure into interchange, rewards cost, balances, finance charges, losses, capital, and servicing cost.

7.3 Implication for the NPV component

The lifetime-value literature should enter the replacement component as a state and duration layer. The component should track whether an account is prospect, booked, activated, active transactor, revolver, promotional borrower, cash-advance user, dormant, delinquent, closed, converted, or charged off. Those states determine which cash flows are possible and how long they are expected to persist. A transactor path mainly converts purchase volume into interchange, rewards, fraud, servicing, and attrition. A revolver path adds finance charges, funding cost, utilization, EAD, and loss. A promo path adds reward liability, teaser-rate economics, funding timing, post-promo decay, and attrition risk.

This state layer is also the practical answer to terminal value. Rather than attach one opaque terminal-value multiple to every account, the component should estimate terminal value conditional on state, vintage, months-on-book, product, relationship depth, risk, macro state, and downstream account management policy. The bank will still need approved terminal-value conventions, but the model should make the behavioral source of terminal value visible. Validation should include vintage holdouts, active/dormant calibration, attrition calibration, and decomposition checks showing whether long-horizon value is driven by durable behavior or by fragile assumptions.

8 Recommended Modeling Architecture

A defensible bank design synthesis is a modular state-transition architecture. The literature motivates the components, while the bank must validate the issuer-specific transitions and cash-flow mappings. Let X_{it} collect customer attributes, internal relationship history, bureau variables, card terms, macro variables, and campaign information. Let Z_{it} denote the latent card state:

$$Z_{it} \in \{\text{prospect, opened, activated, first-use, active, dormant, delinquent, closed, charged-off}\}.$$

A practical model can be written as

$$\Pr(Z_{i,t+1} = z' \mid Z_{it} = z, X_{it}, a), \quad (58)$$

$$\mathbb{E}(S_{i,t+1}, B_{i,t+1}, P_{i,t+1}, D_{i,t+1}, Q_{i,t+1} \mid Z_{i,t+1}, X_{it}, a). \quad (59)$$

Equations (58) and (59) can be implemented as a hurdle model, survival model, dynamic panel, machine-learning model, structural dynamic discrete-choice model, or simulation engine. The choice depends on data scale, interpretability, regulatory constraints, and the decision horizon. The dynamic discrete-choice tradition, exemplified by Rust [44], is useful conceptually because account states and bank actions evolve over time, although a full structural model may be unnecessary for many targeting applications. Dynamic panel methods such as Blundell and Bond [9] provide another reference point when lagged outcomes and persistent customer effects are central.

The architecture should contain the following modules:

1. **Eligibility and offer module.** Forecast approval, line, APR, annual fee, reward terms, disclosures, and compliance constraints.
2. **Causal response module.** Estimate incremental open, activation, first-use, and wallet-share effects of the campaign.
3. **Activation and early-usage module.** Forecast activation, digital-wallet provisioning, first purchase, first 90-day spend, and early dormancy.
4. **Spend and category module.** Forecast purchase volume, transaction count, ticket size, merchant-category mix, and seasonality.
5. **Balance and payment module.** Forecast revolving behavior, promotional balance-transfer behavior, payment rate, and interest revenue.
6. **Risk and loss module.** Forecast delinquency, default, exposure at default, loss given default, fraud, disputes, and recovery.
7. **Attrition and dormancy module.** Forecast inactivity, closure, product switching, retention response, and charge-off exit.
8. **Finance module.** Convert simulated paths into discounted contribution after rewards, servicing, funding, capital, tax/accounting treatment, and acquisition cost.

The output should be a distribution for $\Delta\text{NPV}_i(a)$, not only a point estimate. Marketing decisions should consider expected value, uncertainty, downside tail risk, portfolio concentration, and operational constraints. A portfolio optimizer may rank customers by expected incremental NPV subject to risk, fairness, capacity, and regulatory constraints.

This architecture creates a contract obligation. If a caller asks for NPV, the component should not return a scalar whose source is opaque. It should return the state path, behavioral drivers, financial decomposition, uncertainty, scenario, valuation semantics, and status sufficient to tell whether the value is usable for the declared decision. For example, a line-option score should carry usage, utilization, EAD, expected loss, capital, and sensitivity paths; an existing-client cross-sell score should carry new-card spend and all-bank incremental spend separately; a retention score should carry attrition and concession-cost paths. The operating specification in Section A is therefore not an independent API exercise. It is the engineering form of the model architecture.

9 Component Decomposition for a Bank-Grade Card NPV Engine

The proposed decomposition into losses, balances, pre-provision net revenue (PPNR), and portfolio-management overlays is directionally right. It resembles how banks already organize credit-card forecasting in risk, finance, and stress testing: expected credit loss is commonly decomposed into probability of default (PD), loss given default (LGD), and exposure at default (EAD); account-level balances and usage drive both revenue and exposure; and PPNR converts balances, rates, fees, expenses, and relationship effects into pre-loss operating profit. The Basel credit-risk framework formalizes the PD/LGD/EAD vocabulary for regulatory capital [7]. U.S. stress-testing practice also separates loan losses from PPNR, while forecasting portfolio balances and revenues under macro scenarios [10, 11]. Accounting standards for current expected credit losses require lifetime expected-loss measurement based on historical experience, current conditions, and reasonable and supportable forecasts [24]. These references do not dictate a marketing NPV model, but they support the decomposition as a bank-native control structure.

This decomposition is also the clearest ownership boundary for the replacement component. The component owns the valuation logic that turns behavioral paths into incremental PPNR, expected loss, capital, tax, relationship value, and terminal value. It owns the decomposition labels, assumptions, model versions, lineage, and reconciliation package needed to explain that valuation. It does not own the general ledger, CECL production process, supervisory stress platform, campaign optimizer, underwriting policy engine, account-servicing workflow, rewards fulfillment platform, or enterprise reporting cube. Those systems consume or reconcile component output; they do not become part of the component build.

The classification needs three amendments before it can serve as an NPV optimization architecture. First, it should separate *state variables* from *cash-flow identities*. Purchases, cash advances, payments, lines, promotional balances, delinquency, and inactivity are state or flow variables; interest income, interchange, rewards cost, fraud, servicing, and credit losses are cash-flow consequences of those states. Second, it should separate *forecasts* from *treatment effects*. A high forecast of spend on the new card is not the same as incremental bank spend after cannibalization, as Section 6 emphasized. Every component must therefore produce $Y_i(a) - Y_i(0)$, not only $Y_i(a)$. Third, it should add several missing economic items: interchange and merchant-category economics, rewards and signup-bonus liability, fraud and dispute losses, servicing and collections expense, funding and liquidity cost, capital cost, tax/accounting treatment, fair-lending and adverse-action constraints, and relationship spillovers to deposits or other products.

9.1 Loss Module: PD, LGD, and EAD

The loss block is the most standard part of the proposed taxonomy. Credit scoring and behavioral scoring literatures model the probability that a borrower or account will default or enter delinquency using application, behavioral, bureau, and macro variables [27, 35, 49]. For a card account, a horizon- h expected-loss identity is

$$EL_{i,t,h}(a) = PD_{i,t,h}(a) LGD_{i,t,h}(a) EAD_{i,t,h}(a), \quad (60)$$

with the incremental object

$$\Delta EL_{i,t,h}(a) = \mathbb{E}[EL_{i,t,h}(a) - EL_{i,t,h}(0) \mid \mathcal{I}_{it}]. \quad (61)$$

Equation (60) is an accounting/risk identity, not an independence assumption. In credit cards, PD, LGD, and EAD are linked through utilization, credit line, payment behavior, delinquency state,

collections treatment, and macro conditions. A line increase may raise activation and spend, but it can also raise utilization and EAD. A balance-transfer promotion may raise near-term balances and interest income, but it changes both the timing of EAD and the risk mix of accounts.

The PD component should therefore be a transition or hazard model, not a static score detached from the card lifecycle:

$$\Pr(D_{i,t+h} = 1 \mid Z_{it}, X_{it}, B_{it}, P_{it}, L_{it}, M_{gt}, a), \quad (62)$$

where D is default or charge-off, Z is lifecycle state, B balance, P payment behavior, L line, and M macro state. Months-on-book and vintage should enter this equation because newly originated accounts season differently from mature accounts. That point is operationally important for campaign NPV: an apparently low-loss first-six-month window may simply precede the period in which a vintage reaches peak delinquency.

LGD should be modeled as a recovery and charge-off severity process conditional on default:

$$LGD_{i,t,h} = 1 - \frac{\mathbb{E}[\text{Recoveries}_{i,t:t+h} \mid D_{i,t+h} = 1, \mathcal{I}_{it}]}{\mathbb{E}[EAD_{i,t,h} \mid D_{i,t+h} = 1, \mathcal{I}_{it}]}. \quad (63)$$

The LGD literature emphasizes that recovery rates vary with collateral, seniority, macro state, and workout environment [47]; unsecured credit-card LGD must additionally account for collections strategy, bankruptcy, hardship programs, and sale of charged-off debt. For a marketing NPV engine, LGD is usually less sensitive to the offer than EAD and PD, but it is not constant across macro states or customer segments.

EAD must be tied to the balance and line module:

$$EAD_{i,t,h}(a) = B_{i,t+h}^{\text{principal}}(a) + B_{i,t+h}^{\text{nonprincipal}}(a) + \text{Accrued}_{i,t+h}(a) + \text{UndrawnDraw}_{i,t,h}(a). \quad (64)$$

This is where the user’s proposed “principal versus non-principal balances and losses” belongs. The model should maintain a balance subledger because purchase principal, cash advances, balance transfers, fees, interest, and promotional balances have different APRs, payoff rules, recoverability, revenue recognition, consumer behavior, and regulatory treatment. Rolling them into one balance can make a promotion look profitable by mixing low-margin teaser balances with high-margin revolving balances or by assigning losses to the wrong balance type.

9.2 Balance, Line, and Usage Module

The balance block is the mechanical bridge between usage, revenue, and loss. At monthly frequency, a card account should satisfy a stock-flow identity such as

$$\begin{aligned} B_{i,t+1} = & B_{it} + S_{it}^{\text{purchase}} + S_{it}^{\text{cashadv}} + BT_{it} + Fee_{it} + Int_{it} \\ & - Pay_{it} - Credit_{it} - Chargeoff_{it} - OtherAdj_{it}, \end{aligned} \quad (65)$$

where purchases, cash advances, balance transfers, fees, interest, payments, credits, charge-offs, and adjustments are separately retained. This identity is not optional in a production NPV model: without it, the model can forecast sales volume, balances, interest income, and losses that are mutually inconsistent.

The behavioral forecasting pieces around the identity are:

$$\Pr(Z_{i,t+1} \mid Z_{it}, X_{it}, a), \quad (66)$$

$$\mathbb{E}[S_{ij,t+1}^{\text{purchase}} \mid Z_{i,t+1}, X_{it}, M_{gt}, a], \quad (67)$$

$$\mathbb{E}[Pay_{i,t+1} \mid B_{it}, S_{it}, Z_{it}, X_{it}, M_{gt}, a], \quad (68)$$

$$\mathbb{E}[L_{i,t+1} \mid L_{it}, X_{it}, B_{it}, D_{it}, a], \quad (69)$$

with utilization $Util_{it} = B_{it}/L_{it}$. The household-finance evidence in Section 5 enters (67) and (68); the liquidity and credit-line evidence in Gross and Souleles [26] enters (69); the payment-choice and rewards evidence in Section 6 enters (66) and (67).

The proposed list of balance variables is mostly right but should be amended. “Originations” should be split into application approval, booked account, initial line, activation, and first use. “Sales volume” should be category-level purchase volume, not only total spend, because interchange, rewards, disputes, fraud, and macro cyclicalities vary by merchant category. “Lines” should be modeled as a bank action and constraint, not only a descriptive variable. “Cash advance” should be separated from purchases and balance transfers because it has different fees, APR, risk selection, and customer intent. “Interest fees” should be decomposed into finance charges, late fees, annual fees, balance-transfer fees, cash-advance fees, and fee waivers, especially after the CARD Act evidence that regulation and disclosure alter fee economics [2].

9.3 PPNR and Contribution Module

PPNR is a useful finance lens because it separates operating revenue and expense from credit losses. In a card NPV engine, however, it should be defined at account-month level before aggregation:

$$PPNR_{it}(a) = NII_{it}(a) + NIR_{it}(a) - NIE_{it}(a), \quad (70)$$

where

$$NII_{it}(a) = Int_{it}(a) - FundCost_{it}(a), \quad (71)$$

$$NIR_{it}(a) = Interchange_{it}(a) + FeeIncome_{it}(a) + OtherIncome_{it}(a), \quad (72)$$

$$NIE_{it}(a) = RewardCost_{it}(a) + ServicingCost_{it}(a) + FraudLoss_{it}(a) + DisputeCost_{it}(a) + CollectionCost_{it}(a) + AcqCost_{it}(a). \quad (73)$$

A bank-level stress and planning view is consistent with this separation [10, 11]; the marketing and CLV literatures support discounting future net contribution [8, 45, 53]. For our purposes, the monthly incremental contribution is

$$\Delta\Pi_{it}(a) = \Delta PPNR_{it}(a) - \Delta EL_{it}(a) - \Delta K_{it}(a) - \Delta Tax_{it}(a), \quad (74)$$

and the customer NPV is the discounted sum of (74), plus any one-time acquisition or conversion costs not already embedded in monthly PPNR.

Deposits should be included, but not as an undifferentiated PPNR line item. For a card-only campaign, deposits are a relationship spillover:

$$\begin{aligned} \Delta RelValue_{it}(a) = & \Delta DepositMargin_{it}(a) + \Delta LoanMargin_{it}^{other}(a) \\ & + \Delta FeeIncome_{it}^{other}(a) - \Delta Servicing_{it}^{other}(a). \end{aligned} \quad (75)$$

This term should be estimated only when the campaign plausibly changes relationship depth, primary-bank status, or deposit behavior. Otherwise, adding deposit PPNR to card NPV risks attributing organic household banking value to a card offer. In the existing-client setting, randomized holdouts are especially important because relationship value is strongly confounded by pre-existing engagement.

9.4 Cost-Cutting and Portfolio-Management Overlays

Cost-cutting and portfolio-management topics are not separate objectives from NPV; they are control and segmentation layers around the same component engine. Inactivity/attrition should be a state-transition model,

$$\Pr(Z_{i,t+1} \in \{\text{dormant}, \text{closed}\} \mid Z_{it}, X_{it}, S_{it}, B_{it}, \text{Service}_{it}, a), \quad (76)$$

building on duration and customer-base models [15, 20, 46]. Its NPV role is to shorten or lengthen the cash-flow horizon and to trigger retention, line-management, or closure actions. Cost-cutting based on inactivity is only valid if it does not destroy positive future option value or create adverse selection in the remaining portfolio.

Promotional accounts should be represented by explicit promo states:

$$\text{PromoState}_{it} = (\text{APR}_{it}^{\text{promo}}, \text{expiry}_{it}, B_{it}^{\text{promo}}, \text{reward liability}_{it}). \quad (77)$$

This is necessary because signup bonuses, teaser APRs, balance transfers, and deferred-interest offers create cash-flow timing that differs from ordinary revolving balances. A promo account can have high early sales volume and negative NPV if reward and funding costs arrive before durable profitable balances or if post-promo attrition is high.

Product conversion should be modeled as a treatment-induced transition among products:

$$\Pr(\text{Product}_{i,t+1} = p' \mid \text{Product}_{it} = p, Z_{it}, X_{it}, a), \quad (78)$$

with the incremental value measured against organic conversion. This is the same causal problem as new-card issuance: the bank should not credit the conversion campaign for customers who would have converted anyway, and it should subtract cannibalized annual fees, rewards economics, or portfolio balances from the old product.

Months-on-book and vintage should be required dimensions, not optional reporting labels. They enter PD, spend, payment, attrition, and profitability because accounts season after origination, campaigns are launched in different macro states, and underwriting or pricing policies change across cohorts. A production reporting cube should therefore support segmentation by at least product, channel, campaign, offer terms, vintage, months-on-book, FICO or risk band, geography, macro state, line/utilization band, revolver/transactor status, principal/non-principal balance type, promotional status, relationship depth, and protected-class proxy monitoring where legally appropriate. Flexible segmentation is a reporting and governance requirement; it should not imply that every segment receives a separate unconstrained model. Hierarchical or pooled models are often preferable when segment-level data are sparse.

9.5 Schematic System Design

Figure 2 shows the intended system architecture. The important feature is that customer state, bank action, and macro state feed all behavioral modules; the finance layer then imposes accounting identities and converts simulated paths into incremental cash flows. The reporting layer is downstream of the model, so segmentation can be flexible without changing the definition of NPV.

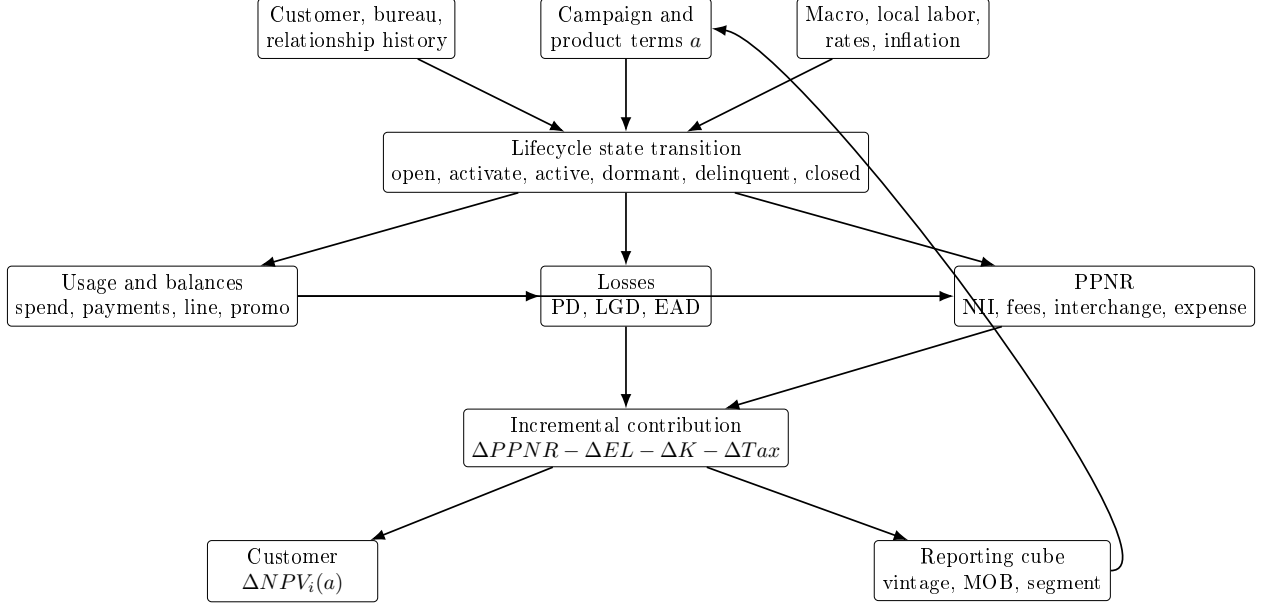


Figure 2: Schematic architecture for a component-based incremental NPV engine.

9.6 Connection Back to Incremental NPV

The component architecture is useful only if every module feeds the same counterfactual cash-flow primitive introduced in Equation (4). The final object should be

$$\Delta NPV_i(a; d, s, \pi) = -C_i^{\text{acq}}(a) + \sum_{h=0}^H \delta_h \mathbb{E}[\Delta CF_{i,t+h}(a, \pi; s) \mid \mathcal{I}_{id}] + \delta_H \mathbb{E}[\Delta TV_{i,t+H}(a, \pi; s) \mid \mathcal{I}_{id}], \quad (79)$$

where each Δ is relative to no offer, no conversion, or the relevant business-as-usual action. The module-level PPNR, expected-loss, capital, tax, and relationship-value terms enter through ΔCF ; acquisition cost and terminal value are shown separately because they are often governed by different finance assumptions. Table 3 summarizes the handoff from each component to Equation (79).

This framing also clarifies what should be amended in the original classification. Losses, balances, and PPNR are essential, but the production system should explicitly include rewards, interchange, fraud/disputes, servicing and collections, capital and funding, taxes/accounting, relationship spillovers, and causal cannibalization. The model should preserve the principal/non-principal subledger and vintage/months-on-book dimensions because they affect both economics and governance. Finally, cost-cutting decisions should be evaluated as actions inside the same NPV system rather than as external heuristics; an inactive account, a promotional account, or a product conversion is valuable only to the extent that its incremental discounted contribution is positive after losses, costs, and cannibalization.

10 Decision and Measurement Pitfalls in Card NPV

Several practical concerns about NPV are legitimate. They are not objections to using NPV; they are reasons the NPV system must be explicitly staged, causal, and uncertainty-aware. The marketing and CLV literatures motivate a discounted contribution objective [8, 16, 45, 53], but the card setting

Table 3: Component outputs required by the incremental NPV model.

Component	Required output	Connection to NPV
Response and activation	$\Delta \text{Pr}(\text{open})$, $\Delta \text{Pr}(\text{active})$, $\Delta \text{Pr}(\text{first use})$	Gates all future spend, balance, expense, and loss paths.
Usage and balances	Incremental purchases, cash advances, balance transfers, payments, lines, promo balances, and utilization	Drives interchange, rewards, finance charges, funding cost, EAD, and capital.
Losses	ΔPD , ΔLGD , ΔEAD , and ΔEL	Subtracted from PPNR as expected credit loss; also informs capital and downside-risk constraints.
PPNR	Incremental net interest income, non-interest income, and non-interest expense	Provides pre-loss contribution before expected loss, capital, tax, and relationship spillovers.
Attrition and conversion	Incremental dormancy, closure, retention, and product-migration paths	Determines cash-flow duration and whether value is new, retained, or cannibalized from another bank product.
Relationship module	Incremental deposit margin and other-product contribution	Added only when credibly caused by the card action, preferably with randomized or quasi-experimental evidence.
Reporting segmentation	Aggregation by product, vintage, MOB, risk, geography, promo, and balance type	Supports governance, monitoring, and cost actions; does not by itself define the optimization target.

adds long-horizon credit, funding, capital, and account management dynamics. The right response is not to replace NPV with response, approval, or sales volume. It is to make clear which NPV is being estimated, for which decision, on which population, and under which downstream policy.

10.1 NPV Is a Long-Horizon Latent Quantity

Customer NPV is unintuitive because it is not directly observed. The bank observes partial cash flows over finite calendar time; it does not observe the counterfactual no-offer path, the customer’s full outside-wallet behavior, or the terminal value of the relationship. A more honest notation is therefore

$$\Delta \text{NPV}_i(a; \theta, \pi) = \sum_{h=0}^H \delta_h \mathbb{E}_\theta[\Delta CF_{i,t+h}(a, \pi) \mid \mathcal{I}_{it}] + \delta_H \mathbb{E}_\theta[\Delta TV_{i,t+H}(a, \pi) \mid \mathcal{I}_{it}], \quad (80)$$

where θ collects model parameters and scenario assumptions, π is the future account-management policy, ΔCF is the one-period incremental cash-flow primitive from Equation (4), and ΔTV is terminal value. This notation makes the instability visible: rankings can change when assumptions about spend decay, revolve rate, attrition, default timing, recovery, funding cost, capital cost, or terminal value change.

A bank implementation should therefore report not only $\widehat{\Delta \text{NPV}}_i(a)$, but also a sensitivity or

posterior distribution:

$$\left\{ \Delta \text{NPV}_i(a; \theta^{(s)}, \pi^{(s)}) \right\}_{s=1}^S. \quad (81)$$

For campaign selection, the robust decision statistic may be expected incremental NPV, lower-tail NPV, probability of negative NPV, or expected NPV subject to capital and loss constraints. This is consistent with the stress-test view that balance, revenue, and loss paths should be evaluated under macro scenarios [10, 11]. It also prevents proxy metrics such as first-90-day spend from silently becoming promotion criteria for lifetime value.

10.2 Card Value Is Behaviorally Heterogeneous

The second issue is that card profitability is not one-dimensional. Two customers with the same first-month spend can have opposite economics if one is a low-risk transactor with high interchange and modest rewards cost, while the other revolves promotional balances, pays slowly, uses cash advances, and later charges off. The appropriate object is a heterogeneous treatment effect over the full cash-flow vector:

$$\tau_V(x, a) = \mathbb{E}[V_i(a) - V_i(0) \mid X_i = x], \quad (82)$$

where V_i is not one outcome but the discounted contribution generated by spend, balances, payments, losses, rewards, fraud, servicing, funding, and capital. Uplift and causal-forest methods are useful for estimating heterogeneous treatment effects [18, 36, 54], and statistical treatment-rule work emphasizes choosing actions by expected welfare or payoff under heterogeneity rather than by average effect alone [39].

This implies that the model should not rank customers by a single revenue or risk proxy. A useful diagnostic decomposition is

$$\begin{aligned} \tau_V(x, a) = & \tau_{\text{interchange}}(x, a) + \tau_{\text{finance}}(x, a) + \tau_{\text{fees}}(x, a) - \tau_{\text{rewards}}(x, a) \\ & - \tau_{\text{servicing}}(x, a) - \tau_{\text{fraud}}(x, a) - \tau_{\text{loss}}(x, a) - \tau_{\text{funding/capital}}(x, a). \end{aligned} \quad (83)$$

The decomposition need not be estimated by separate independent models; indeed, the components are linked through balances and states. But it should be reported, because a high average NPV segment can hide subsegments whose value comes from fragile assumptions about revolve, funding, or losses.

10.3 The Evaluated Population Changes Through the Funnel

The population changes as the customer moves from prospect to applicant to approved account to booked account to active user. This is a selection problem: the covariates observed, the eligible action set, and the conditional value distribution all change after each gate. The sample-selection literature is the classical warning that outcome models estimated only on selected observations need not describe the original population [28]. In the card funnel, the same issue appears in operational form:

$$\mathcal{P}_0 = \{i : \text{marketable prospects}\}, \quad (84)$$

$$\mathcal{P}_1 = \{i \in \mathcal{P}_0 : \text{respond or apply}\}, \quad (85)$$

$$\mathcal{P}_2 = \{i \in \mathcal{P}_1 : \text{approved and line assigned}\}, \quad (86)$$

$$\mathcal{P}_3 = \{i \in \mathcal{P}_2 : \text{booked and activated}\}. \quad (87)$$

The right estimand changes with the conditioning set. For example,

$$\tau^{\text{target}}(x) = \mathbb{E}[\text{NPV}_i(\text{contact}) - \text{NPV}_i(\text{suppress}) \mid i \in \mathcal{P}_0, X_i = x], \quad (88)$$

$$\tau^{\text{approve}}(x) = \mathbb{E}[\text{NPV}_i(\text{approve}) - \text{NPV}_i(\text{decline}) \mid i \in \mathcal{P}_1, X_i = x], \quad (89)$$

$$\tau^{\text{usage}}(x) = \mathbb{E}[\text{NPV}_i(\text{line, terms}) - \text{NPV}_i(\text{baseline}) \mid i \in \mathcal{P}_2, X_i = x]. \quad (90)$$

These are not interchangeable. A targeting model can use prospect and relationship data but not application-only variables. An underwriting model can use application and bureau data that were unavailable at prospect selection. An active-account management model can use payment, utilization, transaction, service, and delinquency history that do not exist for non-booked prospects. Therefore, a bank should maintain separate model cards, validation populations, and data-availability timestamps for each funnel stage.

10.4 The Counterfactual and Objective Change by Decision

The fourth issue is that “the NPV objective” is not one universal treatment contrast. It is a family of decision-specific contrasts. Direct-marketing selection asks whether contact has positive incremental value net of contact cost [16]. Portfolio budget allocation asks whether card marketing beats redeploying the same budget to another product, channel, or retention program. Underwriting asks whether approval creates value relative to decline, subject to credit policy, fairness, compliance, and risk appetite. Line assignment asks which line maximizes value over a feasible menu while respecting loss and capital constraints. Formally,

$$a_d^*(x) \in \arg \max_{a \in \mathcal{A}_d(x)} \mathbb{E}[U_d\{\Delta \text{NPV}_i(a), \text{Risk}_i(a), \text{Cap}_i(a), \text{Fair}_i(a)\} \mid X_i = x, \mathcal{P}_d], \quad (91)$$

where d indexes the decision stage and $\mathcal{A}_d(x)$ is the feasible action set at that stage. The utility U_d may be expected NPV for low-risk targeting, constrained NPV for underwriting, or portfolio NPV subject to budget, capacity, capital, and fairness constraints.

Table 4 gives the operational version. The main governance point is that a model validated for one row should not be reused for another row without re-estimating the counterfactual and checking that the conditioning population and action set match.

10.5 Decisions Are Interdependent Over Time

Finally, decisions are interdependent. Acquisition changes the population that underwriting sees. Underwriting and line assignment change the balance, utilization, and risk states that account management later sees. Future line, pricing, collections, retention, or conversion policies change the lifetime economics assumed at acquisition. The dynamic treatment-regime and Markov decision-process literatures provide useful language for this problem: the value of an action depends on the future policy that follows it [40, 42, 44].

A compact representation is

$$Z_{i,t+1} \sim p_\theta(Z_{i,t+1} \mid Z_{it}, X_{it}, a_t), \quad (92)$$

$$CF_{it} = c_\theta(Z_{it}, X_{it}, a_t), \quad (93)$$

$$\pi_t : (Z_{it}, X_{it}) \mapsto a_t, \quad (94)$$

with policy value

$$V_i^\pi(z, x) = \mathbb{E}_\theta^\pi \left[\sum_{h=0}^H \delta_h CF_{i,t+h} + \delta_H TV_{i,t+H} \mid Z_{it} = z, X_{it} = x \right]. \quad (95)$$

Table 4: Decision-specific counterfactuals for card NPV.

Decision	Population	Counterfactual	Objective
Targeting	Marketable prospects or existing clients	Contact versus suppress	Incremental NPV net of campaign cost.
Budget allocation	Eligible campaign universe	Card campaign versus redeploy budget	Portfolio NPV per scarce marketing dollar or capacity unit.
Underwriting	Applicants or responders with application data	Approve versus decline, or approve with terms versus decline	Risk-adjusted NPV subject to policy and compliance constraints.
Line assignment	Approved or preapproved accounts	Line amount ℓ versus alternative line amounts	NPV over line menu, accounting for usage, EAD, losses, and capital.
Account management	Booked accounts	Maintain, reprice, retain, convert, line change, or close	Forward NPV under updated behavior and risk state.

The acquisition model should therefore state its assumed downstream policy π : for example, line-increase rules, repricing rules, collections policy, retention offers, dormancy closure, and product-conversion policy. If those policies change, the upstream acquisition NPV should be recalibrated. This is especially important for cost-cutting initiatives: closing inactive accounts, reducing promotional offers, tightening lines, or converting products may improve near-term expense or losses while reducing the value of cohorts that were acquired under older assumptions.

The practical recommendation is to treat card NPV as a policy-evaluation system, not a one-time score. Each major decision should store the population, available variables, action set, counterfactual, downstream policy assumption, cash-flow decomposition, and sensitivity range. That metadata is not bureaucracy; it is what prevents an upstream model from optimizing a downstream world that the bank no longer operates.

11 Empirical Design and Validation

11.1 Data requirements

A bank-grade data set should join campaign exposure, channel, creative, offer terms, application, underwriting, activation, card transactions, balances, payments, delinquency, charge-off, recoveries, fraud, disputes, servicing, relationship products, deposits, payroll indicators, bureau refreshes, and local macro series. The data should preserve timing. Variables measured after campaign exposure must not be used as if they were available at targeting time.

For the existing-client new-card problem, the data must also identify cannibalization. At minimum, the bank should measure whether new-card spend comes from:

- a. genuinely incremental customer spend;

- b. spend shifted from a competitor card;
- c. spend shifted from the bank’s existing card;
- d. debit or deposit activity shifted onto credit;
- e. temporary promotion-induced spend that decays after the offer.

The first two categories are the cleanest direct sources of new issuing-bank card value, holding other relationship effects fixed. Debit or deposit substitution, internal old-card shift, and promotion timing can be positive, neutral, or negative depending on interchange, rewards, deposit economics, funding, losses, servicing, and durability.

The data set should also identify the evidence grade for outside-wallet and spend-source estimates. Directly observed or internally anchored quantities include old-bank-card spend, new-card spend, bank-observed debit activity, balance transfers into the new card, payoffs from identified accounts, card-on-file changes visible to the bank, and post-promo decay on the bank’s own book. Externally benchmarked quantities may use bureau tradeline attributes, network or merchant aggregates where available, public macro and credit series, and licensed market data. Model-imputed or scenario-only outside-wallet quantities should be retained for sensitivity analysis, but they should not be reported as captured competitor wallet or used as proof of incremental NPV.

11.2 Promotion criteria

The primary promotion criterion should be randomized or credibly quasi-experimental lift in incremental NPV. Response AUC, approval AUC, activation lift, first-purchase lift, and first-90-day spend are useful diagnostics, but they are proxy metrics. They should nominate models for further testing rather than establish lifetime NPV validity.

Recommended validation splits include:

- time-based holdouts for economic drift;
- campaign randomized holdouts for treatment effects;
- vintage holdouts for lifetime trajectories;
- macro-stress backtests around unemployment, rate, and inflation shocks;
- segment calibration across score bands, income proxies, geography, tenure, relationship depth, channel, and product;
- counterfactual cannibalization analysis for existing-client offers;
- validation anchors for outside-wallet and movable-wallet estimates, including balance-transfer or payoff records, old-card spend decay, debit/deposit substitution, card-on-file migration, bureau aggregates where available, account-aggregation samples, and post-promotion decay.

These validation designs should map directly to component gates. A spend model that fails macro-drift validation should not be used for scenario-sensitive campaign ranking. A causal lift model that lacks overlap or holdout evidence should not be used to claim incremental NPV, even if it predicts responders well. A booked-account model that uses post-approval payment history should not be used for prospect targeting. A decomposition that does not reconcile to finance and risk views should not be used for production ranking until the assumption difference is explained and approved.

An outside-wallet model that has no validation anchor should not be used for precise dollar capture claims. A target-population diagnostic should compare prospects, contacted customers, responders, applicants, approved accounts, booked accounts, activated accounts, and active incumbents before a model trained on one population is used for another. These are veto conditions for component use, not academic footnotes.

12 Identification, Endogeneity, and Data Strategy

The proposal so far decomposes card value into modules: spend, usage, balances, payments, losses, PPNR, costs, capital, and relationship effects. That decomposition is necessary, but it creates a serious identification problem. Many of the variables that predict NPV are also variables that the bank controls, targets, or observes only after a selected customer passes through the funnel. A model can predict observed portfolio outcomes under the current policy and still be wrong for the decision question, “what is the incremental NPV if we take action a instead of baseline action a_0 ?”

The causal-inference literature provides the language for this distinction. Let $Y_i(a)$ denote the future cash-flow path, or one component of it, that would occur if customer or account i received action a . The component needs objects such as

$$\mathbb{E}[Y_i(a) - Y_i(a_0) \mid X_i, d, s, \pi], \quad (96)$$

not merely $\mathbb{E}[Y_i \mid T_i = a, X_i]$. The latter is an observed-outcome prediction under the historical treatment rule. It equals the causal object only under an identification argument: randomized assignment, conditional exchangeability with overlap, a valid instrument or encouragement design, a valid discontinuity, a credible difference-in-differences design, or another defensible quasi-experiment [3, 29–31, 43]. For a production NPV component, this is not a philosophical distinction. It determines whether the output may be used to rank actions or only to monitor expected outcomes.

12.1 Where endogeneity enters the card NPV modules

Endogeneity first appears at marketing selection. Campaigns are usually not sent randomly. Customers with better scores, deeper relationships, attractive deposit signals, prior engagement, lower expected loss, or lower contact cost may be more likely to receive an offer. If the model compares contacted customers with uncontacted customers without accounting for this assignment rule, it will credit the campaign for value that was already present. A response model has the same problem in a different form: responders are not a random subset of prospects, and their post-response value does not identify the value of contacting the marginal prospect.

The second problem is funnel selection. The evaluated population changes from prospect to applicant, applicant to approved account, approved account to booked account, booked account to activated account, and activated account to seasoned account. At each stage, variables, available outcomes, and economic meaning change. A model trained on active accounts conditions on response, approval, booking, and activation. It may be useful for servicing an active portfolio, but it does not by itself estimate prospect NPV. The sample-selection problem described by Heckman [28] is therefore not a side issue; it is central to using the component across the funnel.

The third problem is endogenous product terms. Credit line, APR, promotional duration, balance-transfer terms, rewards, signup bonus, fee waivers, and retention offers are assigned by policies that already use risk, expected profitability, capacity, customer tenure, or channel. Those terms then change usage, balances, payments, losses, rewards cost, funding cost, and attrition. Naively regressing NPV on line or APR will often recover a mixture of the customer selection rule

and the causal effect of the term. In particular, high-line accounts can look safer because the bank gave high lines to safer customers, while a line increase for a marginal account can still raise EAD, utilization, and loss.

The fourth problem is simultaneity among balances, payments, utilization, and losses. Payments reduce balances, balances generate interest and funding cost, line affects utilization, utilization affects risk scores, delinquency affects collections treatment, and collections treatment affects recovery and attrition. Conditioning on variables that lie after the action, such as early utilization or first-payment behavior, can be appropriate for an account-management decision made later, but it is leakage for a prospect-targeting or initial-line decision. Dynamic panel methods such as Arellano and Bond [5] and Blundell and Bond [9] are useful warnings here: lagged outcomes and unobserved heterogeneity cannot be treated as ordinary controls without thinking through the data-generating process.

The fifth problem is endogenous attrition, dormancy, and account-management policy. An account can become inactive because the customer lost interest, the bank reduced rewards, the line was cut, service quality deteriorated, fraud or dispute experience changed the relationship, another card became more attractive, or the customer entered financial distress. Closure, retention, repricing, hardship, and conversion policies are themselves actions. If the component estimates lifetime value only among surviving accounts, it builds in survivorship bias. If it ignores future policy, it values a customer under an operating regime that may not be the one the bank will actually run.

The sixth problem is macro and geography confounding. Market conditions, employment, rates, inflation, housing, merchant mix, and local shocks affect spend, payments, balances, and credit losses. But campaigns and policy changes also occur in calendar time and may be geographically targeted. A campaign run in a strong labor market can appear profitable because unemployment was falling, not because the offer had high incremental value. A line-tightening policy implemented during a downturn can look harmful or helpful depending on which macro controls and comparison cohorts are used. This is why the macro literature in Section 5 should be connected to identification, not used only as a list of predictors.

Finally, relationship value and deposit spillovers are especially vulnerable to confounding. Customers who take a premium card, keep large deposits, use digital channels, and buy other products are often already more engaged. The component should not count deposit growth, brokerage balances, or other-product profit as card-caused NPV unless there is evidence that the card action changed the relationship. Without that evidence, relationship value should be reported as an associated outcome or scenario sensitivity, not added silently to incremental card NPV.

12.2 Identification designs for component modules

The strongest design is randomized treatment assignment with a preserved control group. For marketing and cross-sell, this means campaign holdouts, channel holdouts, offer-cell randomization, and, where feasible, randomization over creative, rewards, promo duration, or line invitation. The component should treat those experiments as first-class data assets. It should ingest the randomization unit, assignment probability, treatment arm, eligibility rule, holdout flag, campaign calendar, and offer terms. When a randomized design is valid and sufficiently powered, the treatment effect can be estimated directly or with covariate adjustment, and heterogeneous treatment-effect methods can be used to learn which customers have positive incremental NPV.

Randomization is not always feasible. Encouragement designs can still help when the bank can randomize an invitation, reminder, channel placement, or prequalified offer while the final take-up decision remains voluntary. The instrumental-variables literature, especially the local-average-treatment-effect framework of Angrist et al. [4], is relevant because the effect is identified for cus-

tomers whose treatment status is changed by the encouragement, not for every customer. The NPV component should therefore label such estimates as complier effects and should not extrapolate them to customers with very different take-up behavior without a separate model-risk approval.

When assignment is observational but rich pre-treatment data are available, propensity-score and doubly robust methods are useful starting points [31, 43]. The required component discipline is pre-treatment feature construction, overlap checking, balance diagnostics, and explicit logging of the estimand. Doubly robust or double/debiased machine-learning estimators can reduce sensitivity to flexible nuisance models [18], and causal forests can estimate heterogeneous effects [54]. These methods do not make selection vanish. They require that the variables sufficient to make treatment as-good-as-random are observed, measured before treatment, and available for both treated and control populations.

Regression-discontinuity designs can be useful around score cutoffs, credit policy thresholds, line-assignment bands, prescreen eligibility thresholds, or pricing tiers [34]. The strength of the design is local comparison near a threshold; the weakness is that the effect may not generalize far from that threshold. If the component uses such evidence, the response should state the threshold, bandwidth, population, and extrapolation status. For line assignment, this is often a better design than comparing all high-line and low-line accounts.

Difference-in-differences and event-study designs can be useful for policy changes, rewards changes, pricing changes, market shocks, or state-level implementation differences. They require a credible comparison group and pre-trend evidence. Modern difference-in-differences work shows that staggered treatment timing and heterogeneous effects can make naive two-way fixed-effect event studies misleading [17, 48]. The component should therefore record the cohort timing, comparison population, pre-trend diagnostics, and estimator class. A failed pre-trend or placebo test should downgrade the estimate to diagnostic status.

For outcomes observed only after selection, the component should separate the selection equation from the outcome equation. A prospect-level model may need application probability, approval probability, booking probability, activation probability, and conditional lifetime value as separate components. Selection corrections in the spirit of Heckman [28] can be useful when there is a defensible exclusion restriction. More often, the practical remedy is data preservation: keep suppressed prospects, nonresponders, applicants, declines, approvals, bookings, inactive accounts, closures, and charge-offs in the analytic base, with the decision-time information available at each stage.

For sequential account-management decisions, static treatment effects are not enough. A card acquired today may later receive line increases, repricing, retention offers, hardship treatment, collections treatment, or product conversion. Dynamic treatment-regime and Markov decision-process literature [40, 42] supports the requirement that the component tag the downstream policy bundle used in valuation. If the bank changes the downstream policy, the acquisition NPV should not be assumed unchanged.

The result is a hierarchy of evidence for NPV use. Randomized experiments with preserved controls should support production treatment ranking when finance and risk reconciliation also pass. Credible quasi-experiments can support bounded production use within the identified population. Observational adjustment can support challenger models, monitoring, and carefully governed use when overlap and balance diagnostics are strong. Pure prediction should be used for expected outcome forecasting under current policy, not for causal claims about alternative actions.

12.3 Public and licensed external data

External data are valuable, but they should be used for the right job. They support scenario construction, macro controls, benchmarking, vintage context, competitive context, and prior distri-

butions. They generally do not identify the customer-level causal effect of this bank’s card offer, line amount, APR, reward, or retention action. The proposal should state that boundary clearly.

Federal Reserve Enhanced Financial Accounts and Distributional Financial Accounts are useful for household balance-sheet context, distributional wealth and debt aggregates, and macro-financial scenario priors [12]. They can help segment a scenario by household balance-sheet vulnerability or validate whether aggregate portfolio behavior is plausible. They do not reveal how a particular customer would have spent on this bank’s new card under a randomized offer.

Federal Reserve G.19 consumer-credit data provide aggregate revolving and nonrevolving consumer-credit context [13]. Z.1 Financial Accounts provide broader household and financial-sector balance-sheet context [14]. These series are useful for macro controls, scenario alignment, portfolio benchmarking, and explaining whether a backtest period was credit-expansionary or credit-contractionary.

The New York Fed Household Debt and Credit Report and the underlying Consumer Credit Panel ecosystem provide benchmark information on balances, delinquency, credit-card debt, and consumer-credit transitions at aggregate and segmented levels [22]. The useful role for the NPV component is external benchmarking of delinquency, credit-card balance, and transition assumptions, especially by vintage, age, score band, or geography where public cuts are available. The role is not replacement of issuer-level PD, LGD, EAD, or activation calibration.

Labor-market and regional-economic data should feed the spend, payment, and loss modules as scenario and control variables. BLS Local Area Unemployment Statistics can provide local unemployment and labor-force measures [51]. BEA regional accounts can provide state and local personal income, GDP, and related regional indicators [50]. Census ACS can provide demographic, income, housing, commuting, and household composition controls at aggregate geographies [52]. These data are useful for macro and geography controls, segmentation, monitoring, and fairness-risk review subject to legal and compliance governance. They should be joined using approved geography and time keys, not used to infer prohibited characteristics or make unsupported individual causal claims.

CFPB public credit-card data, including card agreements, terms surveys, and related public consumer-finance information, can help benchmark product terms, fees, APR structures, and market practices [19]. These sources are useful for competitive and policy context, especially when the component compares offer terms or explains why rewards, fees, and disclosure changes should be explicit. They are not a substitute for the bank’s own reward cost, activation, revolve, loss, and attrition experience.

FRED and ALFRED are useful data-delivery and vintage-control sources [23]. ALFRED-style real-time vintages matter because a backtest that uses revised unemployment, income, inflation, or rate data can overstate the information the bank actually had at campaign or line-decision time. The NPV component should preserve both the event date and the macro-data vintage used in training and scoring.

Because the bank has a Refinitiv/LSEG license, the component team should also inventory entitled LSEG Workspace and economic-data feeds [37, 38]. Depending on entitlement, useful categories may include rates and yield curves, inflation and macro series, market-implied recession or credit-risk indicators, consumer-sector equity and earnings data, issuer and competitor financials, credit spreads, asset-backed securities market indicators, and merchant or sector indicators. These feeds can improve funding-cost curves, discounting, stress scenarios, competitive context, and early-warning monitoring. The proposal should not assume that any particular proprietary field is licensed or approved for model use until data governance confirms entitlement, permitted use, lineage, retention, and redistribution limits.

12.4 Internal data and selection-safe construction

The bank’s internal data are the only plausible source for issuer-specific activation, usage, cannibalization, loss response, reward cost, servicing cost, and relationship spillover. The component should therefore require an analytic base table that preserves the funnel rather than only the booked portfolio. At minimum, it should include campaign eligibility, suppression rules, contact assignment, randomization and holdout flags, channel, creative, offer terms, prescreen variables, application data, approval, decline, refer, override, booking, activation, transaction, balance, payment, delinquency, charge-off, recovery, fraud, dispute, servicing, complaints, rewards, account-management treatments, deposits, payroll indicators, relationship products, bureau refreshes, and local macro joins.

The most important engineering requirement is time. Each row used for training or scoring should have an as-of timestamp, decision timestamp, feature-cut timestamp, outcome window, treatment timestamp, macro-data vintage, and policy-bundle identifier. Variables created after treatment should be available only for later decisions. For example, first-purchase behavior can help value a retention offer after activation, but it cannot be used to decide whether to target the prospect before the card is opened. This feature-cut rule is the practical defense against post-treatment leakage.

For existing-client cross-sell, the internal data should preserve all-bank spend, old-card spend, new-card spend, debit activity, deposit flows, merchant category, digital-wallet provisioning, autopay, promotional balances, balance-transfer activity, cash advances, and reward-redemption behavior. The component should estimate not only new-card spend but all-bank incremental value. If the bank cannot observe competitor-card spend, the proposal should state the missing counterfactual and use experiments, bureau aggregates where available, or sensitivity bounds rather than pretending the outside-card counterfactual is observed.

For losses, internal data must connect underwriting, line assignment, utilization, delinquency, collections, charge-off, recovery, hardship, bankruptcy, and account closure. PD, LGD, and EAD should be estimated on decision-appropriate populations. A booked-account PD model can support booked portfolio forecasting; prospect-level expected loss additionally needs application, approval, booking, and activation selection. Principal and non-principal balances, promotional balances, fees, accrued interest, cash advances, and balance transfers should be kept separate because they have different revenue, loss, payment, and accounting behavior.

For PPNR and costs, internal finance data should provide rewards expense, interchange, fees, interest income, charge reversals, fraud expense, collections cost, servicing cost, dispute cost, acquisition cost, funding transfer pricing, capital charge, tax convention, and allocated overhead. The component should reconcile behavioral forecasts to finance definitions, but it should not become the official ledger. Differences between model cash flows and finance reporting should be explained through valuation-semantics identifiers and reconciliation reports.

The component should also preserve decision logs from current systems. The historical policy that generated treatment assignment is itself data. Marketing scores, eligibility rules, prescreen criteria, underwriting score cutoffs, manual overrides, line caps, APR matrices, channel capacity, budget constraints, collection queues, retention rules, and closure rules help explain why a customer received an action. Without those logs, observational adjustment is often weak because the most important confounder is the bank’s own prior view of the customer.

12.5 Estimation ledgers and evidence grades

The replacement component should maintain estimation ledgers that match the objects it claims to estimate. These ledgers are not separate governance paperwork; they are the way the component

prevents outside wallet, movable wallet, causal response, and NPV from being collapsed into one opaque score.

The selection and funnel ledger should compare suppressed prospects, contacted prospects, responders, applicants, declined applications, approvals, booked accounts, activated accounts, dormant accounts, and active incumbents. It should record target-population weights, overlap diagnostics, and segment calibration. If the model is trained on booked or active accounts but used on prospects, the response should expose that transportability status.

The offer and terms ledger should record contact channel, creative, reward, signup bonus, annual fee, APR, balance-transfer terms, promotional duration, line, fee waivers, repricing, retention treatment, eligibility rule, and any manual or policy override. It should also record whether each term was randomized, policy-driven, negotiated, manually overridden, or imputed. This ledger is essential because line, APR, rewards, and promotions are bank actions, not passive covariates.

The outside-wallet and cannibalization ledger should record the source of each outside spend, outside balance, movable-wallet, and spend-source estimate: directly observed, internally anchored, externally benchmarked, experimentally identified, quasi-experimental, model-imputed, or scenario-only. It should also record the uncertainty interval and any cap or reconciliation constraint, such as estimated moved spend not exceeding plausible outside eligible spend.

The dynamic behavior ledger should decompose observed changes into total spend or borrowing growth, share movement to the bank, internal cannibalization, debit or deposit substitution, promotion timing, balance build, payment response, runoff, and attrition. This ledger should be validated out of time, by vintage, and by stress or macro regime.

The causal evidence ledger should record the experiment, holdout, encouragement, cutoff, policy change, or observational adjustment supporting each treatment-response estimate. It should include overlap, balance, pre-trend, placebo, effective-sample-size, and sensitivity diagnostics. If the evidence is only associational, the output should be labeled as predictive or diagnostic, not causal incremental NPV.

The downstream-policy ledger should record the assumed future line, repricing, retention, collections, closure, hardship, and product-conversion policies. A customer acquired under one assumed future policy can have a different value under another. The NPV output should therefore carry the policy bundle identifier and should be revalidated when that bundle changes.

12.6 Production implications for the NPV component

The identification strategy should be visible in the component contract. Each NPV response should carry an identification design identifier, population stage, decision context, treatment definition, baseline definition, feature-cut timestamp, macro-data vintage, downstream-policy bundle, experiment or holdout identifier when available, overlap or support score, causal-evidence level, outside-wallet evidence grade, movable-wallet evidence grade, cannibalization evidence grade, target-population diagnostic, and allowed-use status. This metadata is not decoration. It tells the caller whether the number is an incremental value estimate, an expected-outcome forecast, a scenario diagnostic, or an extrapolation requiring approval.

The component should implement veto diagnostics. It should warn or refuse production action-ranking use when treatment-control overlap is poor, balance diagnostics fail, randomization is broken, pre-trends fail, placebo effects are large, post-treatment leakage is detected, the model is trained on booked or active accounts but called for prospects, macro variables use revised rather than as-of vintages, the caller asks for a decision outside the approved population, CATE estimates are unstable, or finance/risk reconciliation fails. The component should also warn when estimated moved wallet exceeds plausible outside wallet, when temporary promotion lift is being used as durable wal-

let share, or when a model-imputed outside-wallet value is being used as if it were directly observed or experimentally identified. These gates are how the literature becomes an operating control.

The component should also distinguish three output classes. First, *causal incremental NPV* is supported by randomized or credible quasi-experimental evidence and may be used for governed action ranking within the approved population. Second, *policy-conditional expected NPV* predicts outcomes under the historical or declared policy and may be used for planning, monitoring, and portfolio forecasting, but not as proof that a different action would create value. Third, *diagnostic or benchmark NPV* uses external data, weak observational evidence, or scenario assumptions and should be labeled for sensitivity analysis or management review. A single scalar without this class label is not an acceptable replacement component.

These requirements should be part of the replacement component package. Each module should have an identification note, a data-lineage specification, an experiment and holdout inventory, a decision-stage analytic-base design, a leakage-control checklist, a public and licensed data inventory, and a validation gate that separates predictive accuracy from causal incremental value. These artifacts determine which submodel outputs can enter Equation (1) for each caller and which outputs must remain warnings, diagnostics, or assumptions.

13 Open Gaps and Reviewer Risks

The public literature does not fully identify issuer-specific new-card usage and incremental NPV. Payment-choice and rewards studies show mechanisms, but the activation probability for a particular bank, product, offer, channel, and customer segment is a proprietary empirical object. Likewise, CLV models provide powerful baselines for transaction frequency and duration, but bank NPV requires integration with expected credit loss, rewards, fraud, funding, capital, and regulatory constraints.

The largest reviewer risks are:

1. treating activation or first spend as lifetime NPV evidence;
2. treating raw propensity as causal lift;
3. ignoring cannibalization from the bank's existing cards;
4. estimating spend without macro and employment state dependence;
5. separating marketing CLV from credit risk and balance-sheet cost;
6. using public rewards estimates as universal program parameters;
7. reusing one NPV model across targeting, underwriting, line assignment, and account management without changing the counterfactual;
8. validating on booked or active accounts while applying the result to prospects or applicants without correcting for funnel selection;
9. treating product terms such as line, APR, promo duration, rewards, or fee waiver as exogenous predictors when they were assigned by bank policy;
10. using public macro, credit, or market data as if they identify customer-level treatment effects rather than scenario, control, benchmark, or prior information;

11. adding deposit, relationship, or other-product value to card NPV without evidence that the card action caused the relationship change;
12. treating observed new-card spend as captured outside wallet;
13. treating estimated outside wallet as offer-specific movable wallet;
14. treating wallet size or movable-wallet size as NPV before rewards, losses, funding, capital, fraud, servicing, and attrition are netted;
15. using model-imputed wallet estimates as if they were directly observed or experimentally identified;
16. treating temporary promotion lift as durable wallet-share capture.

14 Alternatives and Tradeoffs

Several simpler approaches are possible, and each is useful for a limited purpose. The reason for recommending a modular NPV component is not that these alternatives are worthless. It is that none of them is sufficient as the replacement valuation component for a large bank once the decision spans marketing, new-card cross-sell, line assignment, offer design, account management, finance reconciliation, and model-risk review.

A response or propensity model is the natural first alternative. It is fast to build, easy to monitor, and operationally familiar. It can improve campaign execution by estimating who is likely to respond, apply, activate, or make a first purchase. The direct-marketing literature, however, already warns against optimizing response alone: the decision should depend on expected net return, not merely on response probability [16]. In a card portfolio, this limitation is sharper because response does not price rewards, losses, funding, capital, servicing, fraud, attrition, or cannibalized spend. Propensity models should therefore remain funnel modules and diagnostics, not the final value object.

A second alternative is a standard CLV model with a finance overlay. This is closer to the economic target because CLV explicitly discounts future contribution and has mature models for purchase frequency, duration, and retention [8, 20, 45, 46, 53]. It is a useful baseline for lifetime activity. The tradeoff is that plain CLV does not by itself solve the card problems that matter most for bank use: PD, LGD, EAD, line and APR policy, principal versus non-principal balances, rewards liability, funding transfer pricing, capital charges, and within-bank cannibalization. A finance overlay can patch some of these terms, but if the overlay is added after behavior is forecast, it can miss the fact that product terms and account-management policies change behavior itself.

A third alternative is to rely on randomized holdouts and causal lift studies without a full dynamic NPV engine. This is attractive because experiments are the cleanest evidence for incremental value, and the proposal should treat campaign holdouts and offer-cell randomization as first-class assets. The limitation is horizon and policy dependence. A campaign experiment may identify 90-day activation, spend lift, or early loss differences, but it does not automatically identify five-year value under future line increases, repricing, retention offers, collections policies, macro shocks, and terminal value assumptions. Experiments should anchor the causal layer; they do not remove the need for a governed forecast and valuation layer.

A fourth alternative is a monolithic machine-learning value score trained on historical realized profitability. This can produce a strong offline fit and may be useful as a challenger benchmark. It is risky as the production replacement if it hides the target population, treatment assignment, feature timing, finance definitions, and decomposition. A model trained on booked or active accounts can

look accurate while being invalid for prospects. A score that uses post-treatment features can leak future information. A realized profitability target can combine organic customer quality with the effect of the bank action. These are not cosmetic model-risk issues; they change the counterfactual being estimated.

The recommended architecture is therefore a modular NPV component with clear valuation semantics. It uses response, activation, CLV, credit-risk, PPNR, capital, cost, and causal-lift models, but it does not collapse them into an uninterpreted scalar. Its cost is additional data lineage, governance, and reconciliation work. Its benefit is that a reviewer can inspect why a customer is valuable, whether the value is causal or predictive, which population the evidence supports, how losses and capital enter, and what assumptions would change the ranking. The practical rollout should still be phased: start with a controlled batch acquisition or prescreen slice, retain simpler models as baselines and diagnostics, and expand use cases only when the component-level evidence and integration contracts support them.

15 Using the Component in Existing Bank Decision Processes

The component should be used only through an explicit decision-context contract. That contract is simple in concept even if the implementation schema is detailed. The caller must say who or what is being scored, which decision is being valued, what action alternatives are feasible, what baseline defines the counterfactual, what information was available at the decision time, what valuation semantics and scenario are being used, and what downstream policy is assumed. The component then returns a decomposed value object, status, warning set, and lineage. The caller remains responsible for action choice and execution.

The most important use cases are different enough that they should not share one generic interpretation of NPV. For new-to-bank targeting, the baseline is usually suppress and the action is contact through a channel and offer. For existing-client new-card cross-sell, the baseline is no new-card offer and the central risk is cannibalizing existing bank-card or debit activity. For underwriting, the baseline is decline or refer and the action must remain inside approved credit policy. For line assignment, the component values a menu of feasible line or term options and must expose usage, EAD, loss, and capital for each option. For retention, inactivity, product conversion, and promotional treatment, the baseline is the organic account path and the action may change attrition, rewards, servicing, balances, and losses.

Valuation semantics must be visible for every use. A value number should not move through the bank unless its horizon, discounting, funding, capital, tax, cost allocation, rewards, relationship value, cannibalization, terminal value, scenario, and downstream-policy assumptions are tagged. This is not a technical nicety. Without the tag, two callers can compare numbers that are both called NPV but are economically different.

The response should be decomposed enough to diagnose why a customer or option is valuable. A high expected value from interchange is different from a high expected value from revolve. A high value dominated by terminal value is different from one dominated by near-term spend. A positive new-card-spend lift with near-zero all-bank-spend lift is a cannibalization warning, not a success. A high-line option with higher spend but materially higher EAD and capital should not be treated as a pure revenue win. These distinctions are why the component must return paths, decomposition, uncertainty, and status, not only a scalar.

Status semantics protect callers from silent misuse. A no-score because the use case is unapproved is different from a technical timeout. A degraded score because merchant-category history is missing is different from an out-of-domain population. A batch job can succeed while individual

rows fail; an option grid can score one line amount while rejecting another. Those states should be explicit in the response and preserved for replay.

16 Migration, Governance, and First Production Slice

The replacement should be introduced as a controlled component migration, not as a one-time model launch. The first production slice should remain narrow: one batch acquisition or prescreen consumer, one offer family, one valuation semantics bundle, frozen feature snapshots, campaign holdouts, and shadow scoring against the current component. This slice is valuable because it exercises the full NPV chain without adding real-time underwriting or account-management risk.

Migration should begin by inventorying the current component. The bank needs to know every consumer, field, threshold, fallback, owner, latency class, report, and downstream decision that currently depends on the old value. It then needs a valuation-definition comparison: horizon, discounting, funding, capital, tax, cost, reward, terminal value, scenario, and policy assumptions in the old component versus the new one. Without this comparison, a ranking change can look like a model improvement when it is actually an assumption change.

Shadow scoring should compare both values and decisions. The bank should measure rank correlation, top-decile overlap, segment-level aggregate NPV, component decomposition tie-outs, no-score and degraded-score rates, disagreement patterns, and downstream contract behavior. Material disagreements should be put in a register and classified as expected improvements, assumption differences, data issues, unsupported use cases, or blockers. Cutover should occur by controlled consumer and decision context, not globally.

Governance should separate five questions. Does the model predict the behavioral paths it claims to predict? Does it estimate causal lift for actions that change outcomes? Does the decomposition reconcile to finance, treasury, risk, and product definitions? Do the data and integration pipelines preserve as-of timing, schema stability, replay, and status semantics? Is the proposed use case approved by business, model risk, compliance, legal, and technology owners where required? A positive answer for marketing targeting does not approve underwriting, line assignment, account closure, collections, adverse action, CECL, stress testing, or official finance reporting.

Rollback should be designed before launch. Trigger examples include schema breaks, no-score or degraded-score rates above threshold, failed finance or risk tie-out, material unexplained ranking movement, replay failure, feature leakage, out-of-domain drift, or use outside the approved decision context. Rollback may mean returning to the legacy component, suppressing a campaign, using a conservative no-score policy, or routing to manual review, depending on the consumer and use case.

17 Conclusion

The replacement component should estimate incremental, risk-adjusted customer NPV, not response, approval, activation, or new-card spend alone. The literature supports the mechanisms that make this necessary: market conditions, employment, liquidity, credit terms, rewards, adoption, and customer duration affect spend, balances, payments, attrition, and losses. The literature does not supply this bank's activation rate, cannibalization rate, movable-wallet share, loss elasticity, funding cost, capital charge, or relationship spillover. Those are issuer-specific empirical quantities that must be estimated from the bank's own experiments, quasi-experiments, internal data, external benchmarks, and finance/risk reconciliations.

The proposal therefore recommends a modular valuation component with explicit decision context, valuation semantics, scenario, downstream-policy assumption, decomposition, status, lineage,

and allowed-use metadata. The first slice should be a controlled batch acquisition or prescreen use case, with holdouts, shadow scoring, reconciliation against the current component, and governed cutover. Broader use in underwriting, line assignment, existing-account treatment, finance planning, or account management should be approved only after the relevant counterfactual, population, data lineage, and validation evidence support that use.

A Reference Operating Specification for the Replacement NPV Component

The preceding sections establish what the component must estimate and why. This section translates those requirements into the contract by which existing bank systems call the replacement component. These contracts are not generic platform design. They are controls against the exact failure modes identified above: NPV is latent, value is heterogeneous, populations change through the funnel, counterfactuals differ by decision, and downstream policies change upstream economics. The purpose is to make the NPV component replaceable, testable, auditable, and useful across many callers without letting one ambiguous value field leak into incompatible decisions.

The operating specification therefore has a narrow job. It tells each caller how to declare the decision context, baseline, feasible action set, feature stage, scenario, valuation semantics, and downstream policy assumption. It tells the component how to return value, decomposition, sensitivity, warnings, status, and lineage. It does not tell marketing which customers to contact, underwriting which applications to approve, account management which treatment to execute, or finance which official ledger number to book.

A.1 Decision-context contract

Every request must declare the decision context, baseline action, candidate action or option grid, population, feature stage, scenario, valuation semantics, and downstream policy assumption. This is not administrative ornament. It is the only way to prevent a value model validated for contact-versus-suppress from being reused for approve-versus-decline, line assignment, or account closure.

The component should support at least three groups of consumers.

Acquisition and new-card cross-sell. These calls occur before the account is booked or before a new product is accepted. The available features may be prospect-level, prescreen-level, relationship-level, or application-level. The primary risk is using post-decision data or mistaking new-card spend for incremental all-bank value.

Table 5: Decision-context contract for acquisition and new-card cross-sell.

Context	Caller	Scored unit	Baseline and action set	Required output
New-to-bank targeting	Marketing platform	Prospect or household	Suppress versus contact through one or more channels/offers	Incremental NPV net of contact and acquisition cost; caller owns campaign execution.

Context	Caller	Scored unit	Baseline and action set	Required output
Existing-client new-card cross-sell	CRM or cross-sell system	Customer, household, or relationship	No new-card offer versus offer menu	Incremental all-bank value after cannibalization, plus diagnostic new-card spend; caller owns offer execution.
Prescreen or preapproval	Prescreen system	Prospect or existing client	No prescreen versus preapproved offer terms	Expected value conditional on prescreen eligibility, response, approval, and booked account economics.
Application underwriting	Underwriting engine	Application	Decline, refer, or approve with feasible terms	Risk-adjusted value under approved credit policy; underwriting owns final decision and compliance process.
Initial line and offer assignment	Line or offer engine	Approved application or account-action pair	Feasible line/terms menu	Option-level NPV, usage, EAD, loss, capital, and sensitivity paths; caller owns chosen line.

Existing-account treatment. These calls occur after an account exists. The component may use payment, transaction, utilization, delinquency, service, dispute, rewards, and dormancy history that is unavailable earlier in the funnel. That means validation on active accounts cannot be back-projected to prospects without a separate selection argument.

Table 6: Decision-context contract for existing-account treatment.

Context	Caller	Scored unit	Baseline and action set	Required output
Line increase or decrease	Account-management or line engine	Account-action pair	Current line versus feasible line menu	Forward NPV, utilization, EAD, PD, LGD, capital, and attrition effects.

Context	Caller	Scored unit	Baseline and action set	Required output
Retention or inactivity treatment	Retention or servicing system	Account or relationship	No treatment versus fee waiver, reward, line, closure, or contact option	Incremental retained value net of concession, servicing, and adverse selection.
Product conversion	Product or CRM system	Account-product pair	Organic path versus conversion menu	Incremental value after annual fee, rewards, cannibalization, attrition, and customer-state transitions.
Promotional balance or reward treatment	Offer or balance-transfer engine	Account-action pair	No promotion versus promo terms	Promo-state NPV, reward liability, funding, post-promo attrition, and loss path.
Collections-sensitive forward value	Servicing or collections system where permitted	Account	Current treatment path versus approved alternatives	Forward value under hardship, collections, recovery, and compliance constraints; caller owns customer treatment.

Offline finance, risk, and portfolio consumers. Offline consumers should use component outputs for analysis and reconciliation, not as evidence that the NPV component owns finance reporting, CECL, stress testing, or enterprise dashboards.

Table 7: Offline consumers of component output.

Use	Consumer	Scored unit	Boundary
Campaign budget analytics	Marketing analytics	Portfolio or campaign universe	Component supplies value and drivers; budget optimizer remains outside scope.
Vintage profitability review	Product, finance, and risk	Vintage, campaign, product, or segment	Component supplies decomposed values; official finance reporting remains outside scope.

Use	Consumer	Scored unit	Boundary
Scenario and planning analysis	Finance, treasury, risk, planning	Portfolio and segment	Component can evaluate approved scenario bundles; stress and planning platforms remain outside scope.
Model-risk review and monitoring	Model-risk management	Use case, model version, segment	Component supplies lineage, validation, drift, and reconciliation artifacts; model-risk approval remains separate.

A.2 Valuation semantics and assumption contract

NPV is a valuation convention, not a natural constant. Two systems can both claim to use NPV while differing in horizon, terminal value, discount curve, funds-transfer pricing, capital charge, tax treatment, reward liability, cost allocation, relationship value, cannibalization, macro scenario, and downstream policy. Therefore each request and response should carry a canonical `valuation_semantics_id`. That identifier should resolve to a controlled assumption bundle with an owner, approval record, effective date, and version.

Table 8: Valuation semantics required for a decision-grade NPV output.

Choice	Required definition	Primary approver	Risk if unspecified
Value basis	Incremental versus total value; pre-tax versus post-tax value	Finance and product	Callers compare incompatible scores.
Horizon and terminal value	Explicit horizon, terminal-value formula, and decay assumptions	Finance, product, model risk	Rankings become dominated by hidden long-horizon assumptions.
Discounting and funding	Discount curve, funds-transfer-pricing curve, liquidity basis	Treasury and finance	Balance-heavy accounts are over- or under-valued.
Capital	Economic, regulatory, stress, or business capital charge	Risk capital and finance	High-loss or high-EAD growth is ranked too favorably.
Costs and rewards	Acquisition, servicing, rewards, fraud, disputes, collections, and allocated versus marginal cost rules	Finance, product, rewards finance	High spend is mistaken for profitable value.

Choice	Required definition	Primary approver	Risk if unspecified
Relationship value	Whether deposit and other-product value is excluded, included, or reported separately	Product, finance, relationship analytics	Card offer receives credit for organic household value.
Cannibalization	Old bank card, debit, deposit, outside-card, and organic path treatment	Product and causal validation	New-card spend is mistaken for bank-incremental spend.
Scenario and policy	Macro scenario, rate path, loss scenario, and downstream account-management policy π	Risk, finance, policy owners	Upstream value assumes a future operating world the bank will not run.

A.3 Request contract

The production request schema should be machine-readable and contract-tested. The human proposal should specify the semantic obligations, not every eventual JSON field. Each field must have a business definition, type, unit or currency where applicable, cardinality, allowed values, requiredness by decision context and operating mode, and backward-compatibility rule.

Table 9: Minimum request contract for the NPV component.

Field group	Business definition	Requiredness	Compatibility rule
Request envelope	request_id , trace_id , caller, use case, schema version, model version, request timestamp, feature-cut timestamp	Required	New optional fields may be added; changed meaning requires schema version increment.
Scored unit	Person, household, relationship, application, account, or account-action pair identifiers and join keys	Required	Identifier semantics must be stable and lineage-preserving.
Decision context	Context, baseline action, candidate action, action grid, action owner, feature stage, and permitted action set	Required	New contexts require model-risk and consumer approval.
Product terms	Line, APRs, fees, rewards, signup bonus, promo terms, disclosures, balance transfer and cash-advance terms	Conditional	Missing terms must trigger no-score, degraded-score, or approved defaults.

Field group	Business definition	Requiredness	Compatibility rule
Customer state	Bureau, internal relationship, deposits, payroll indicators where permitted, transaction history, payments, balances, utilization, service, fraud, disputes, delinquency, and closure state	Context-specific	Training-serving availability must be enforced by feature-cut timestamp.
Scenario and assumptions	Macro, unemployment, rates, inflation, valuation semantics, downstream policy assumption, and scenario version	Required	Assumption versions must remain replayable.

The component should reject, warn, or degrade gracefully when callers submit inputs inconsistent with the declared context. A prospect call should not silently use booked-account payment history. A line-assignment call should not score a line outside the approved feasible set. A batch request with mixed feature availability should encode row-level statuses rather than hide no-score rows in aggregate output.

A.4 Response contract and decomposition

The response should be a valuation object with decomposition and lineage. The component should not only return **npv**. At minimum it should return the expected value, horizon values, uncertainty or sensitivity, component cash-flow paths, behavioral paths, status, warnings, and replay identifiers.

Table 10: Minimum response contract for the NPV component.

Field group	Business definition	Unit	Scope
Response envelope	Request identifiers, schema version, model version, assumption bundle, scenario, timestamp, lineage hash	N/A	Request and row.
Valuation	Expected incremental NPV, total value if approved, horizon values, terminal value, discount basis	Currency	Row and option.
Decomposition	Incremental PPNR, NII, NIR, NIE, expected loss, capital, tax, relationship value, acquisition cost, rewards, fraud, servicing	Currency	Row, option, and horizon.
Behavioral paths	Activation, first use, repeated use, spend, payment, balance, utilization, attrition, PD, LGD, EAD, closure	Probability, currency, ratio	Row, option, horizon.
Uncertainty and sensitivity	Scenario range, confidence or posterior interval where approved, downside metrics, major driver sensitivities	Mixed	Row, option, segment.

Field group	Business definition	Unit	Scope
Status and warnings	Score status, degraded inputs, out-of-domain flags, unavailable features, validation limitations, approved downstream handling	Enum/text	Request, row, option.

The decomposition must sum to the reported NPV within a documented tolerance. If the response includes both new-card spend and all-bank incremental spend, the field names should make the distinction impossible to miss. A report that shows high `new_card_spend_lift` but low `all_bank_incremental_spend_lift` should not be optimized as if the two were the same.

A.5 Score-status semantics

Score status is part of the business contract. A no-score because a use case is unapproved is different from a timeout. A degraded score due to missing merchant-category history is different from an out-of-domain account state. The component should support request-level, row/account-level, and option-level statuses.

Table 11: Score-status semantics and downstream handling.

Status	Trigger	Scope	Downstream handling
Scored	Required inputs, approved use case, in-domain population, successful model execution	Request, row, option	Caller may use value subject to its own policy and approval.
Degraded	Non-critical feature missing, stale, imputed, or lower-quality fallback used	Row or option	Caller may use only if use-case policy permits degraded values; warnings must be logged.
No-score	Required feature missing, unapproved use case, unsupported context, invalid action set, or unavailable valuation semantics	Row, option, sometimes request	Caller must use approved fallback, suppress, or route to manual review; missing value must not be treated as zero NPV.
Out-of-domain	Population, product, macro state, or behavior outside approved model domain	Row or option	Caller should suppress, route to manual review, or use conservative fallback according to policy.
Manual review	Model result requires human or governance review because stakes, conflicts, or policy exceptions exceed thresholds	Row or option	Caller routes according to use-case workflow; component records reason.

Status	Trigger	Scope	Downstream handling
Technical failure	Timeout, service error, schema parse failure, dependency failure	Request or batch partition	Retry or use technical fallback; do not interpret as a governed business no-score.

Mixed-status batch and option-grid responses must be explicit. A batch may succeed at request level while individual rows no-score. An option grid may score one line amount and no-score another because the second is outside the approved action set. Downstream systems must receive these statuses, not only a filtered list of successful scores.

A.6 Operating modes and integration patterns

The component should support three modes. Each mode uses the same valuation semantics but has different latency, feature freshness, idempotency, fallback, and replay requirements.

Table 12: Operating modes for the NPV component.

Mode	Typical consumers	Request shape	Failure and fallback	Audit/replay
Batch scoring	Acquisition, prescreen, campaign planning, shadow scoring, portfolio analytics	Many rows, frozen feature snapshot, one or more offer options	Mixed row statuses; legacy or conservative fallback only if approved	Full input snapshot, model and assumption versions, row lineage, run manifest.
Online scoring	Selected underwriting, servicing, account-management flows	Single customer, account, application, or small option set	Latency timeout, no-score, manual review, or caller-approved fallback	Request/response log, feature-cut timestamp, deterministic replay where possible.
Option-grid scoring	Line, offer, promotion, conversion, treatment menus	One scored unit with feasible action grid	Option-level statuses; invalid options do not invalidate valid options	Option-level lineage, decomposition, policy constraints, and chosen-action record stored by caller.

Idempotency should be defined by request identifier, feature-cut timestamp, model version, schema version, valuation semantics, scenario, and downstream policy assumption. Replay requires retained feature snapshots or reproducible feature queries, retained code artifacts, model artifacts, and assumption bundles. Where exact replay is impossible because a source system cannot recreate

historical inputs, the limitation must be logged and accepted by model-risk and data owners before production use.

B Reference Governance, Migration, and First Production Slice

Migration should be governed as replacement of a production component, not as a greenfield model launch. The bank should run the new component in shadow, compare rankings and decisions against the current component, reconcile aggregate and component economics, analyze disagreements, run downstream contract tests, and define rollback triggers before any production consumer relies on the new value.

B.1 Validation and governance gates

Validation should separate predictive accuracy, causal lift, finance reconciliation, data-quality monitoring, and use-case approval. A model can have good response AUC and still be invalid for NPV if it fails cannibalization, loss, or cost validation. A model can calibrate on booked accounts and still be invalid for prospect targeting if it uses variables not available at targeting time.

Table 13: Governance lenses, owners, veto diagnostics, and reapproval triggers.

Lens	Primary owner	Veto diagnostics	Reapproval trigger
Predictive validation	Model owner and model-risk management	Poor calibration or rank stability for spend, balances, payments, attrition, PD, LGD, EAD, costs, or terminal value	New population, product, model, feature set, or material drift.
Causal/treatment validation	Marketing analytics, experimentation, model risk	Failed holdout lift, weak overlap, unaddressed selection, unstable CATE, cannibalization not identified	New offer, channel, action set, or counterfactual.
Finance reconciliation	Finance, treasury, risk capital	PPNR, rewards, funding, capital, tax, losses, or cost decomposition fails tie-out tolerance	New valuation semantics, accounting policy, FTP curve, capital method, or cost allocation.
Data and integration monitoring	Data engineering, platform, model owner	Missingness, staleness, schema breaks, feature leakage, no-score rate, degraded-score rate, out-of-domain rate	Source-system change, schema change, feature pipeline change, material quality drift.

Lens	Primary owner	Veto diagnostics	Reapproval trigger
Use-case approval	Business owner, compliance, model risk, legal where needed	Unapproved population, action set, fairness concern, adverse-action mismatch, unsupported downstream use	Any new caller, decision context, operating mode, or customer-impacting use.

NPV explanations are not automatically adverse-action reasons or fair-lending explanations. Caller systems that use NPV in credit decisions must maintain their own approved compliance and adverse-action processes.

B.2 Migration artifacts and cutover gates

The first production slice should be a controlled acquisition or prescreen batch use case. The slice should have one consumer, one offer family, one valuation semantics bundle, frozen input snapshots, campaign holdouts, and shadow scoring against the current component. The first release should not try to cover underwriting, line assignment, existing-account treatments, finance analysis, scenario planning, and account management at once.

Table 14: Required migration artifacts before production reliance.

Artifact	Purpose	Owner
Current-consumer inventory	Identify every consumer, field, threshold, fallback, report, owner, and latency class that depends on the current component	Product, technology, model owner.
Legacy schema and output map	Map old fields to new fields, compatibility fields, retired fields, and downstream contract changes	Data engineering and consumer owners.
Valuation-definition comparison	Compare old and new horizon, discounting, funding, capital, tax, cost, reward, terminal-value, scenario, and policy assumptions	Finance, risk, model owner.
Shadow-score run manifest	Preserve data snapshot, command or job, model versions, assumption bundle, scenario, random seeds where relevant, and output paths	Model owner and platform.
Ranking and decision-impact comparison	Quantify rank correlation, top-decile overlap, decision movement, budget movement, and segment impacts	Business analytics and model owner.
Segment aggregate tie-out	Compare aggregate NPV and drivers by product, channel, vintage, months-on-book, risk band, geography, and relationship depth	Finance, risk, product.

Artifact	Purpose	Owner
Decomposition tie-out	Verify PPNR, losses, funding, capital, rewards, fraud, servicing, acquisition cost, tax, and relationship value sum to NPV	Finance and model owner.
Disagreement register	Document material old-versus-new disagreements and whether they are expected improvements, data issues, assumption differences, or blockers	Model owner, business, model risk.
Downstream contract tests	Confirm schema validation, mixed statuses, option-level results, fallback, replay, and rollback interface	Platform and consumer system owners.
Rollback plan	Define triggers, owner, interface, timing, and data consequences for reverting to legacy or conservative fallback	Business owner, technology, model risk.

Cutover criteria should be numerical and use-case specific. They should include acceptable segment-level aggregate tie-out tolerances, decomposition residual tolerance, ranking and decision-impact thresholds, disagreement-rate thresholds and required disposition, contract-test pass requirements, no-score and degraded-score thresholds, and named rollback triggers. This proposal does not hard-code the tolerances because the right values depend on the legacy component, action stakes, population size, and regulatory sensitivity.

B.3 What this proposal does not approve

This proposal does not by itself approve any production use case beyond a shadow-tested first slice. It does not approve reuse of the score in underwriting, line assignment, account closure, collections, adverse-action reasoning, official finance reporting, CECL, or stress testing. Each such use requires its own decision-context contract, valuation semantics, validation evidence, governance approval, compliance review where applicable, and consumer-system integration test.

C Source-Support Audit

This paper is a structured literature survey for product and model-design purposes. It is not yet a complete model-governance literature audit. Primary source pages, abstracts, and working-paper summaries were inspected on June 26–27, 2026. Full-text technical sections, appendices, citation-count metadata, retraction checks, and full backward/forward snowballing remain open. Table 16 records how the main sources are used.

The claim-support rule for future model work should be simple: if a statement is used to justify a model requirement, it should be backed either by checked technical literature, by bank data or an experiment, or by an explicit design convention. Literature can justify the mechanism; internal bank evidence must justify issuer-specific rates, elasticities, and cannibalization; design conventions can justify output tags, status semantics, and contract fields. Anything else should be labeled as assumption or open gap, not quietly promoted to fact.

The wallet-movement claims in this proposal are deliberately treated as project derivations and data requirements. Public literature and payment-choice evidence motivate adoption, usage, rewards, and treatment-response mechanisms, but they do not observe this issuer’s competitor-card

wallet or identify the offer-specific movable fraction. Those quantities require internal anchors, experiments, quasi-experiments, or explicitly labeled scenario assumptions.

Table 15 records the claim-support discipline that the production model package should make more granular. It is framed by claim rather than by source because reviewers will press on conclusions: why the component is incremental, why activation is separate from use, why cannibalization matters, why PD/LGD/EAD belong in marketing value, why valuation semantics are necessary, and why the component should not own caller actions.

Table 15: Claim-support ledger for the proposal’s main design claims.

Design claim	Support class	What remains to be proven or approved
Campaign selection should optimize incremental value, not response alone.	Literature support from direct-marketing profit targeting, CLV, and causal treatment-effect methods; project synthesis in Equations (2)–(5).	Bank must validate incremental NPV with holdouts or credible quasi-experiments for each use case.
Spend, payments, balances, and losses are state dependent.	Literature support from unemployment, rebate, liquidity, credit-card limit/APR, CARD Act, and macro-shock evidence.	Bank must estimate issuer-specific elasticities and monitor macro, employment, and liquidity drift.
New-card issuance, activation, first use, repeated use, and wallet share are distinct states.	Literature support from payment-choice and rewards studies; project derivation in adoption/use and state-transition equations.	Bank must estimate transition probabilities and treatment effects by product, channel, segment, and feature stage.
Treatment-response modules require an identification design before they can support incremental NPV.	Causal-inference, program-evaluation, IV, RD, DiD, double/debiased ML, and uplift literatures support the distinction between observed-outcome prediction and causal treatment effects.	Bank must document randomized, quasi-experimental, or observational identification evidence, overlap, balance, leakage controls, and allowed use for each module and decision context.
Existing-client new-card value requires cannibalization measurement.	Project derivation in Equations (42)–(51); causal/uplift methods support identification strategy.	Bank must estimate all-bank incremental spend, old-card shift, debit shift, and uncertain outside-card/new-consumption components.

Design claim	Support class	What remains to be proven or approved
Observed new-card spend must be decomposed before it enters NPV.	Project derivation in Equations (39)–(41).	Bank must validate incremental consumption, competitor-card shift, old-bank-card cannibalization, debit/deposit substitution, promotion timing, and measurement uncertainty with internal data, anchors, or experiments.
Outside card wallet, movable wallet, causal response, and NPV are distinct objects.	Project derivation in Equation (40) and decision-context logic in Equation (1).	Bank must tag evidence grade and uncertainty for outside-wallet and movable-wallet estimates; large outside wallet should not be promoted to captured wallet or positive NPV without causal and financial validation.
Lifetime card value requires state and duration modeling beyond plain CLV.	CLV, customer-base, and duration literature support active-state, frequency, monetary, and attrition components.	Bank must validate transactor, revolver, promo, dormant, conversion, and charge-off state paths and terminal-value sensitivity.
Losses should be decomposed into PD, LGD, and EAD linked to balances and lines.	Regulatory, credit-risk, and accounting sources support PD/LGD/EAD and lifetime expected-loss framing.	Bank must validate calibration, line response, vintage seasoning, principal/non-principal subledgers, and stress behavior.
PPNR, rewards, funding, capital, tax, costs, and relationship spillovers must be explicit.	Supervisory PPNR/stress sources, CLV contribution logic, and project cash-flow identities support the decomposition.	Finance, treasury, risk, product, and model-risk owners must approve semantics and reconciliation tolerances.
NPV must carry valuation semantics and downstream-policy assumptions.	Project derivation from latent long-horizon NPV and dynamic policy-value literature.	Owners must approve horizon, discounting, funding, capital, tax, terminal value, relationship value, scenario, and policy-bundle conventions.
The component should value options but not execute caller actions.	Design convention and governance boundary supported by decision-specific counterfactual analysis.	Each caller must maintain its own action policy, compliance process, adverse-action process where applicable, and integration approval.

Design claim	Support class	What remains to be proven or approved
External public and licensed data should support scenarios, controls, benchmarks, and priors, not replace internal causal evidence.	Official public-data and LSEG/Refinitiv-source descriptions support their role as external macro, credit, market, and product-term data sources.	Bank must confirm entitlements, usage rights, lineage, real-time vintages, join keys, and whether each use is scenario, control, benchmark, prior, or causal evidence.

Table 16: First-pass source-support ledger.

Source	Classification	Use in this paper
Bult and Wansbeek [16]	Direct marketing optimization	Supports the claim that campaign selection is normally an expected-profit problem, not a response-probability problem alone.
Berger and Nasr [8], Rust et al. [45], and Venkatesan and Kumar [53]	CLV and customer-equity foundations	Support the discounted-contribution framing; the paper’s credit-card NPV equation is a bank-specific extension with loss, capital, funding, fraud, rewards, and cannibalization terms.
Ganong and Noel [25]	Household finance	Supports the claim that labor-income and unemployment events affect high-frequency spending and should enter spend forecasts.
Johnson et al. [32]	Income-shock evidence	Supports the claim that liquidity and transfer timing can produce heterogeneous short-run spending responses.
Gross and Souleles [26]	Credit-card account evidence	Supports the claim that credit limits and interest rates affect card debt behavior and interact with liquidity constraints.
Agarwal et al. [2]	Credit-card regulation evidence	Supports the claim that fees, disclosures, and payment nudges are material NPV components and governed levers.
Baker et al. [6]	Macro-shock evidence	Supports the claim that major shocks can alter spending levels and category composition.
Koulayev et al. [33]	Payment-choice model	Supports the distinction between payment-instrument adoption and use.
Agarwal et al. [1]	Rewards and usage evidence	Supports the claim that rewards can shift card spend and debt; not used as a universal activation-rate estimate.
Prelec and Simester [41]	Behavioral mechanism	Supports the mechanism that payment method can affect purchase behavior; not used for portfolio calibration.
Schmittlein et al. [46], Fader et al. [20], and Fader et al. [21]	CLV foundations	Support use of transaction-frequency, recency, monetary-value, and active status models as components of card lifetime modeling.

Source	Classification	Use in this paper
Bolton [15]	Duration model	Supports use of relationship-duration or hazard components for attrition and dormancy.
Basel Committee on Banking Supervision [7]	Regulatory capital framework	Supports PD/LGD/EAD terminology and the use of exposure, default probability, and severity as separate risk components; not a marketing NPV optimization paper.
Financial Accounting Standards Board [24]	Accounting standard	Supports lifetime expected-credit-loss framing under current conditions and reasonable forecasts; not used to define customer-level causal lift.
Board of Governors of the Federal Reserve System [10] and Board of Governors of the Federal Reserve System [11]	Supervisory stress-test methodology	Support bank-level separation of projected losses, balances, revenues, and PPNR under macro scenarios; used here as architecture support, not as a customer-acquisition estimator.
Hand and Henley [27], Thomas [49], and Lessmann et al. [35]	Credit-scoring surveys and benchmark	Support use of application and behavioral scoring methods for PD and delinquency/default components; not used to choose one production model.
Schuermann [47]	LGD survey	Supports modeling LGD as a severity/recovery object distinct from PD and EAD, with macro and workout-state dependence.
Rosenbaum and Rubin [43], Lo [36], Wager and Athey [54], and Chernozhukov et al. [18]	Causal/uplift methods	Support methodological framing for treatment effects; not bank-specific evidence.
Imbens and Rubin [30], Angrist and Pischke [3], Imbens and Wooldridge [31], and Hernán and Robins [29]	Causal-inference and program-evaluation foundations	Support the distinction between potential outcomes, observed outcomes, identification assumptions, overlap, and estimands; used to define component evidence levels, not to supply card-specific parameters.
Angrist et al. [4] and Lee and Lemieux [34]	Quasi-experimental identification methods	Support use of encouragement/instrumental-variable and local-threshold designs when valid; the bank must document the instrument, threshold, exclusion restriction, bandwidth, and population before production use.
Callaway and Sant’Anna [17] and Sun and Abraham [48]	Difference-in-differences and event-study methods	Support caution about staggered timing and heterogeneous treatment effects; used to require pre-trend, cohort, and estimator diagnostics for policy changes.

Source	Classification		Use in this paper
Heckman [28]	Sample-selection foundation		Supports the warning that models estimated after response, approval, or booking need not describe the original prospect population.
Arellano and Bond [5] and Blundell and Bond [9]	Dynamic methods	panel	Support caution about lagged outcomes, unobserved heterogeneity, and dynamic feedback in balance, payment, and loss models.
Manski [39]	Statistical treatment rules	treat-	Supports choosing actions by expected payoff under heterogeneous effects, rather than ranking by a single average response or risk measure.
Puterman [42], Murphy [40], and Rust [44]	Dynamic and methods	decision policy-value	Support representing acquisition, underwriting, line assignment, and account management as linked sequential decisions whose value depends on downstream policy.
Board of Governors of the Federal Reserve System [12], Board of Governors of the Federal Reserve System [13], Board of Governors of the Federal Reserve System [14], and Federal Reserve Bank of New York [22]	Public financial and consumer-credit data	macro-	Support external scenario construction, portfolio benchmarking, and credit cycle context; not used as issuer-level activation, spend, PD, LGD, EAD, or causal treatment-effect evidence.
U.S. Bureau of Labor Statistics [51], U.S. Bureau of Economic Analysis [50], U.S. Census Bureau [52], and Federal Reserve Bank of St. Louis [23]	Public market, demographic, and macro-vintage data	labor-regional, and	Support geography-time controls, stress/scenario variables, segmentation, monitoring, and real-time-vintage backtesting; usage must respect legal, compliance, and data-governance constraints.
Consumer Financial Protection Bureau [19]	Public credit-card product data		Supports competitive and product-term context for APRs, fees, agreements, and market practices; not a substitute for bank-specific profitability or response estimates.
London Stock Exchange Group [38] and London Stock Exchange Group [37]	Licensed macro, and financial data sources	market,	Support a candidate inventory of rates, curves, macro, credit, issuer, competitor, sector, and market indicators, subject to entitlement, lineage, permitted-use, and retention review.

No paper in this survey is used to conclude a universal activation rate, spend elasticity, or NPV parameter. The activation and cannibalization parameters for a large U.S. retail bank should be estimated from internal experiments or credible quasi-experiments.

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